From Parkinson's disease genome-wide association signals to testable mechanistic hypotheses

An integrated GWAS2Mech analysis

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Abstract

Background: Genome-wide association studies identify loci associated with Parkinson's disease, but linkage disequilibrium and context-dependent gene regulation complicate the assignment of causal variants, effector genes, and downstream biological processes. We developed GWAS2Mech to integrate statistical fine-mapping with functional annotation, molecular quantitative trait locus (QTL) colocalisation, external benchmarking, pathway and protein-association analyses, and public perturbation data.

Methods: Seventeen European-ancestry GWAS datasets for Parkinson's disease and related phenotypes were harmonised to GRCh38 and analysed with a 1000 Genomes 30× European linkage disequilibrium reference. Associated loci underwent SuSiE-RSS fine-mapping, variant annotation, and molecular-QTL colocalisation. Candidate genes were compared with Open Targets, tested for pathway and protein-association enrichment, and linked to source-matched single-cell perturbation signatures. Each final record retained its study, locus, variant, molecular phenotype, pathway, experimental context, and unresolved evidence fields.

Results: Eight studies yielded 69 nominal 95% credible sets. Cross-layer quality control supported 24 candidate genes for downstream analysis. Public perturbation resources contained usable contrasts for eight candidates, and 14 gene–pathway–experiment records passed the exploratory gene-set permutation filter. Only one record linked the same genetically prioritised pathway gene to the corresponding perturbation response: *NSF* and Golgi vesicle transport. *KANSL1* perturbations produced respiration-, mitochondrial-, and cytoskeleton-related transcriptional signatures, but these terms lacked same-gene pathway-driver support in the genetic enrichment analysis.

Conclusions: GWAS2Mech converts heterogeneous association evidence into traceable, testable mechanistic hypotheses while preserving uncertainty across evidence layers. The analysis prioritised *NSF*-related vesicular transport for experimental follow-up, but did not establish allele-to-pathway causal direction, a clinically validated mechanism, or a therapeutic effect.

Keywords: Parkinson's disease; genome-wide association study; fine-mapping; molecular QTL; colocalisation; pathway analysis; protein association; genetic perturbation

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1 Introduction

Genome-wide association studies (GWAS) have identified reproducible Parkinson's disease (PD) susceptibility loci, yet most associated variants do not directly reveal the affected gene or biological process. Correlated variants within linkage disequilibrium blocks can support several plausible signals, and regulatory effects may differ by tissue, cell type, transcript, or molecular phenotype. Statistical fine-mapping can reduce the set of candidate variants, but a credible set is a probabilistic summary rather than proof of causality [1, 2].

Mechanistic interpretation therefore requires evidence from several complementary sources. Variant annotations describe predicted molecular consequences and regulatory context; molecular quantitative trait loci (QTLs) connect genetic variation to measured molecular phenotypes; and colocalisation tests whether disease and molecular associations are compatible with a shared signal. Resources such as Open Targets integrate fine-mapping, locus-to-gene, functional-genomic, and disease evidence, providing an external reference for evaluating gene prioritisation [3, 4].

Pathway and protein-association analyses can place prioritised genes within biological systems, but enrichment and network connectivity do not demonstrate causal mediation. Perturbation experiments provide a distinct evidence layer by testing whether changing a candidate gene is accompanied by a transcriptional response in a defined cellular context. Public single-cell perturbation collections improve access to these experiments, although differences in model system, guide design, replication, and preprocessing constrain cross-study interpretation [5, 6].

This study applied GWAS2Mech to 17 European-ancestry GWAS datasets for Parkinson's disease and related phenotypes. The objectives were to identify statistically supported loci and variants, prioritise candidate effector genes through molecular evidence, compare those assignments with Open Targets, and determine which genetically enriched pathways had corresponding public perturbation evidence. The analysis was designed as a provenance-preserving hypothesis-prioritisation framework: every integrated claim remains linked to its source study, locus, molecular phenotype, pathway definition, and experimental context.

2 Materials and methods

2.1 Study design and analytical framework

GWAS2Mech used a sequential evidence-integration design (Figure 1). The workflow began with public GWAS summary statistics and ended with gene–pathway–experiment records classified by the evidence available at each layer. The analysis retained related Parkinson's disease phenotypes as separate studies because case–control disease risk, family-history proxies, medication use, motor subtypes, and treatment responses are not interchangeable outcomes.

2.2 Genetic association processing and fine-mapping

Seventeen harmonised GWAS datasets were obtained from the GWAS Catalog. Variant identifiers, genome coordinates, alleles, association statistics, sample size, and phenotype metadata were standardised and checked before analysis. A GRCh38 European reference panel derived from the 1000 Genomes 30× resource supplied allele-frequency and linkage disequilibrium estimates.

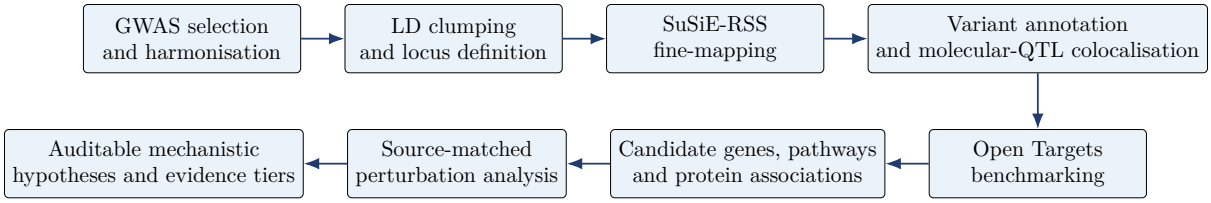


Figure 1: Overview of the GWAS2Mech study design. Each stage retained source identifiers and quality-control fields so that downstream biological hypotheses could be traced to the supporting genetic, molecular, and experimental observations. LD, linkage disequilibrium; QTL, quantitative trait locus.

Genome-wide association signals were clumped into loci, harmonised with the reference panel, and analysed with SuSiE-RSS. The workflow reported posterior inclusion probabilities, nominal 95% credible sets, convergence diagnostics, and unresolved loci.

2.3 Functional annotation and molecular-QTL integration

Fine-mapped variants were annotated with GRCh38-compatible functional resources, including predicted consequences, regulatory annotations, conservation, population frequency, gene constraint, and clinical records. Molecular-QTL analyses then evaluated bulk-tissue, cell-type-specific, and external datasets. Colocalisation support was interpreted as compatibility with a shared statistical signal under the specified model; it was not treated as proof that a named gene was the sole causal effector or that the molecular effect mediated disease risk.

2.4 External benchmarking and systems-level analyses

GWAS2Mech loci, credible sets, genes, and molecular evidence were compared with the Open Targets Genetics release recorded in the resource audit. Candidate genes passing the study's quality-control rules were analysed for pathway over-representation and leave-one-gene-out or leave-one-region-out sensitivity. STRING and BioGRID supplied complementary protein-association context [7, 8]. Network enrichment, direct physical edges, and database-added connector proteins were reported separately to prevent network expansion from being mistaken for additional genetic support.

2.5 Perturbation analysis and evidence integration

Public single-cell perturbation datasets were screened for usable contrasts involving prioritised genes. Within each annotated gene–dataset contrast, target-perturbed cells were compared with the identified control population, and source-matched pathway gene sets were tested by gene-set enrichment analysis [5, 9]. The final integration retained representative GWAS and QTL variants, colocalisation posterior, pathway-driver status, protein-association evidence, normalised enrichment score, experimental model, and missing validation fields. A transcriptional response was not considered a validated disease mechanism unless the genetic gene assignment, molecular direction, target perturbation, and downstream pathway response could be aligned.

2.6 Statistical interpretation and reporting

Thresholds, denominators, and quality-control exclusions were recorded for every analytical stage. Counts of loci, credible sets, genes, pathways, and experimental records represent different units and were not combined as independent discoveries. False-discovery-rate-adjusted pathway statistics and perturbation statistics arose from separate tests. The detailed subsections below state the relevant implementation choices before each result to preserve a complete computational audit. Workflow identifiers follow the executable pipeline; setup, validation, and plotting stages do not receive separate biological-results subsections.

3 Results

3.1 GWAS dataset selection and acquisition (workflow steps 1–2)

3.1.1 Identification and selection of GWAS datasets

Genome-wide association study (GWAS) summary statistics were retrieved from the GWAS Catalog for Parkinson's disease and related phenotypes in populations of European ancestry. The datasets included direct Parkinson's disease case–control associations, family-history-based proxy phenotypes, medication-use phenotypes, Parkinson's disease motor subtypes, and levodopa-induced dyskinesia.

Each GWAS dataset was identified using its unique GWAS Catalog study accession (GCST). Associated publication metadata, phenotype descriptions, ancestry, genome assembly, participant numbers, case–control composition, and available effect estimates were collected to characterise the input studies.

A total of 17 GWAS datasets were downloaded. The datasets were obtained as harmonised summary-statistic files, providing variant-level association statistics rather than individual-level participant genotypes.

The downloaded data represented several distinct phenotype definitions. In particular, family-history associations and medication-use phenotypes were retained as related but non-equivalent measures of Parkinson's disease. Similarly, motor-subtype and treatment-response studies were considered separately from disease-risk GWAS datasets.

Table 1 summarises the study characteristics and sample sizes.

Table 1: Characteristics of the 17 European-ancestry GWAS datasets retrieved for Parkinson's disease and related phenotypes.

GCST accession	Study	Phenotype	Total N	Cases / Controls
GCST90104087	Le Guen (2021)	Parkinson's disease, chromosome X	564,139	16,521 / 547,618
GCST90475830	Verma (2024)	Parkinson's disease	448,956	8,704 / 440,252
GCST90476961	Verma (2024)	Parkinson's disease	315,668	4,047 / 311,621
GCST90104085	Le Guen (2021)	Parkinson's disease, chromosome X	302,908	5,328 / 297,580
GCST90104086	Le Guen (2021)	Parkinson's disease, chromosome X	261,231	10,002 / 251,229

Continued on next page

Table 1 continued from previous page

GCST accession	Study	Phenotype	Total N	Cases / Controls
GCST90319903	Doostparast Torshizi (2024)	Parkinson's disease	161,740	2,864 / 158,876
GCST90473307	UK Biobank WGS Consortium (2025)	ICD-10 G20: Parkinson's disease	458,440	4,217 / 454,223
GCST90043734	Jiang (2021)	Parkinson's disease, PheCode 332	456,348	294 / 456,054
GCST90042681	Jiang (2021)	Maternal history of Parkinson's disease	416,553	6,877 / 409,676
GCST90435905	Zhou (2018)	Parkinson's disease, PheCode 332	396,336	1,127 / 395,209
GCST90042668	Jiang (2021)	Paternal history of Parkinson's disease	393,679	9,960 / 383,719
GCST90079817	Backman (2021)	ICD-10 G20: Parkinson's disease	387,664	1,827 / 385,837
GCST90042694	Jiang (2021)	Sibling history of Parkinson's disease	355,975	1,976 / 353,999
GCST90476962	Verma (2024)	Medication use for Parkinson's disease	315,668	3,573 / 312,095
GCST90281037	Sosero (2024)	Levodopa-induced dyskinesia	4,787	1,612 / 3,175
GCST90000015	Alfradique-Dunham (2021)	Parkinson's motor-subtype quantitative score	3,212	–
GCST90000014	Alfradique-Dunham (2021)	Parkinson's motor-subtype comparison	2,829	–

For quantitative and within-case motor-subtype analyses, conventional disease-case and population-control counts were not applicable or were not fully specified in the extracted metadata. These quantities are therefore not reported as conventional case–control samples.

Importantly, the datasets were not assumed to be statistically independent. Several studies used UK Biobank or Million Veteran Program data, and some datasets evaluated related phenotypes within potentially overlapping cohorts.

3.1.2 Downloaded GWAS variant records

The 17 downloaded datasets collectively contained 269,271,867 variant-association records before quality control. This number represents the sum of records across individual GWAS files and does not represent an equal number of unique genomic variants.

The number of association records differed substantially between studies because of differences in genotyping, imputation, sequencing technologies, variant coverage, and study design.

For example, the UK Biobank whole-genome sequencing GWAS (GCST90473307) contained 84,399,074 initial variant records, whereas each chromosome-X dataset from Le Guen and colleagues contained 209,339 records.

These study-specific differences were retained in the downstream data inventory rather than combining all summary statistics into a single meta-analysis.

3.2 GWAS harmonisation and quality control (workflow step 2)

3.2.1 Harmonisation and quality-control procedures

Quality control was applied independently to each GWAS dataset using GWASLab within the GWAS2Mech processing pipeline.

The objective was to produce consistently formatted variant-association statistics suitable for subsequent reference-panel matching, linkage disequilibrium clumping, and fine-mapping.

The processing procedure included standardisation of chromosome identifiers, genomic positions and allele representations, examination of association statistics, normalisation of supported allele representations, identification of duplicate records, and exclusion of invalid or unsuitable entries.

The quality-control pipeline employed the GWASLab `basic_check` procedure with variant removal, duplicate removal, and allele normalisation enabled.

Variants were subsequently required to have nonmissing P-values within the interval $[0, 1]$. Association statistics were reorganised into a consistent set of columns, including chromosome, position, variant identifier, effect allele, other allele, effect size, standard error, P-value, effect-allele frequency, minor-allele frequency, sample size, imputation information when available, and Z-score.

For binary traits, odds ratios were represented on the log-odds scale where required. Where possible, standardised Z-scores were obtained from the reported effect size and standard error or reconstructed using the available association statistics.

Missing variant-level sample sizes were supplemented using the study-level total sample size when available, with their provenance retained.

Table 2 describes the implemented quality-control procedures.

Table 2: Quality-control operations applied to GWAS summary statistics.

Operation	**Implementation**
Input genome assembly	Harmonised GRCh38 genomic coordinates.
Chromosome and position checks	Invalid or missing chromosome and genomic position information assessed using GWASLab.
Allele checks	Effect and other alleles standardised; supported allele representations normalised.
Association-statistic checks	GWASLab sanity checks applied to available P-values, effect sizes, standard errors, Z-scores, sample sizes, odds ratios, and related statistical fields.
Duplicate handling	Duplicate SNP identifiers, rsIDs, variant-coordinate records, and multiallelic representations examined. The pipeline retained the best-supported record according to its duplicate-resolution procedure.
P-value validity	Missing P-values excluded; remaining values required to satisfy $0 \leq P \leq 1$.
Allele frequency and INFO	Available frequency and imputation-information fields retained and assessed; no additional user-specified global MAF or INFO threshold was imposed at this step.
Sample-size handling	Variant-level sample size retained when available; missing values supplemented from study-level `N_TOTAL` where supported.
GWAS significance threshold	$P < 5 \times 10^{-8}$.
Standardised output	`CHR, POS, SNP, EA, NEA, BETA, SE, P, EAF, MAF, N, INFO, Z`.

3.2.2 Variant retention following quality control

Across the 17 datasets, 266,128,410 of the 269,271,867 initial variant records were retained following quality control, corresponding to a retention rate of approximately 98.83%.

A total of 3,143,457 records were removed (approximately 1.17%).

The largest absolute number of removed records occurred in the whole-genome sequencing dataset GCST90473307, which retained 81,595,123 of its 84,399,074 initial records.

The quality-controlled datasets contained a combined total of 13,148 genome-wide significant variant-association records across the 17 studies.

Table 3 presents the study-specific retention and significance counts.

Table 3: Number of GWAS variant-association records before and after quality control. Genome-wide significance was defined as $P < 5 \times 10^{-8}$.

GWAS accession	Before QC	After QC	Removed	Significant
GCST90104087	209,339	209,339	0	32
GCST90475830	19,670,763	19,603,221	67,542	3,642
GCST90476961	19,607,622	19,540,506	67,116	23
GCST90104085	209,339	209,339	0	32
GCST90104086	209,339	209,339	0	32
GCST90319903	16,692,835	16,668,207	24,628	3,167
GCST90473307	84,399,074	81,595,123	2,803,951	3,246
GCST90043734	11,831,294	11,831,292	2	0
GCST90042681	11,831,277	11,831,275	2	2,966
GCST90435905	27,940,507	27,834,350	106,157	0
GCST90042668	11,831,457	11,831,455	2	1
GCST90079817	393,573	386,728	6,845	0
GCST90042694	11,831,226	11,831,224	2	0
GCST90476962	19,607,622	19,540,506	67,116	2
GCST90281037	5,561,604	5,561,527	77	0
GCST90000015	14,580,738	14,580,728	10	5
GCST90000014	12,864,258	12,864,251	7	0
Total	**269,271,867**	**266,128,410**	**3,143,457**	**13,148**

All 17 datasets were marked as successfully processed by the GWAS2Mech quality-control procedure. However, successful file-level quality control does not imply equivalent statistical power or phenotype validity.

For example, several studies had relatively small numbers of Parkinson's disease cases, and some datasets exhibited unusually low genomic inflation statistics. Such characteristics may influence the number and reliability of detectable associations.

Additionally, three chromosome-X GWAS accessions (GCST90104085, GCST90104086 and GCST90104087) produced identical quality-control association summaries despite having different reported sample sizes. These records were flagged for verification of their source files and should not be interpreted as independent genetic replications without further investigation.

3.3 European linkage disequilibrium reference panel (workflow step 4)

3.3.1 Reference-panel selection

To characterise linkage disequilibrium (LD) between genetic variants, we prepared an ancestry-matched reference panel from the high-coverage 1000 Genomes Project sequencing resource.

The 1000 Genomes 30× resource was selected because it provides phased genetic variants in GRCh38 coordinates, including single-nucleotide variants and insertion/deletion variants, together with structural-variant records represented in the source panel.

The original high-coverage resource comprised 3,202 sequenced individuals. The pipeline used the designated unrelated subset of 2,504 individuals as the starting population resource, from which European-ancestry individuals were selected.

The final European panel contained 503 individuals drawn from five 1000 Genomes populations:

- CEU: 99 individuals
- FIN: 99 individuals
- GBR: 91 individuals
- IBS: 107 individuals
- TSI: 107 individuals

The final panel was constructed separately for the 22 autosomal chromosomes and stored in PLINK 2 genotype-reference format.

The panel contained 70,692,015 variant records across chromosomes 1–22.

Table 4: Characteristics of the prepared European 1000 Genomes 30× LD reference panel.

Characteristic	**Value**
Reference resource	1000 Genomes high-coverage, 30× sequencing
Reference release	NYGC high-coverage 1000 Genomes resource (2022)
Genome assembly	GRCh38
Individuals in source sequencing resource	3,202
Designated unrelated individuals in global resource	2,504
Selected ancestry	European (EUR)
European individuals retained	503
CEU / FIN / GBR	99 / 99 / 91
IBS / TSI	107 / 107
Chromosomes included	Autosomes 1–22
Completed chromosome panels	22 / 22
Variants in final EUR panel	70,692,015
Genotype representation	Phased source VCF converted to PLINK 2 PGEN format
Variant-identity handling	Reference allele, alternate allele, chromosome and position retained for downstream harmonisation; duplicate identifiers resolved during preparation.

3.3.2 Reference-panel processing

Ancestry-specific samples were selected from the source phased variant-call files. Genotype information for the selected European samples was extracted on a chromosome-by-chromosome basis and converted to PLINK 2 reference files.

The pipeline preserved the reference allele and alternate allele representations from the GRCh38 source data. Variant identifiers were retained where available, with coordinate- and allele-based identifiers generated for variants lacking an appropriate identifier.

Duplicate identifiers were checked during reference construction, with inconsistent duplicated entries excluded according to the reference-panel processing configuration.

All 22 autosomal reference panels completed successfully without reported chromosome-level failures.

The final number of variants varied between chromosomes. Chromosome 2 contained 6,088,598 variants, chromosome 1 contained 5,759,060, and chromosome 21 contained 1,002,753.

The total of 70,692,015 refers to variants in the completed European reference panels. A corresponding total number of variants before European subsetting and reference processing was not provided in the current reference statistics and is therefore not reported as a measured filtering result.

3.3.3 Purpose of the LD reference panel

LD represents the non-random correlation between alleles at different genomic positions.

A GWAS may identify many statistically significant variants within the same region because correlated variants can tag the same underlying association.

The European reference panel was therefore used to estimate pairwise variant correlations for LD clumping and subsequent fine-mapping.

For two variants i and j, their pairwise LD correlation was represented using:

$$r_{ij}^2 = \left[\mathrm{Corr}(G_i, G_j)\right]^2 , \tag{1}$$

where G_i and G_j denote genotype dosages for the two variants in the reference population.

The reference panel was used to characterise LD rather than to estimate associations with Parkinson's disease directly.

Chromosome-X associations were not included in this autosomal reference-panel analysis because the prepared panels covered chromosomes 1–22.

3.4 Associated-locus definition by linkage disequilibrium clumping (workflow step 5)

3.4.1 Reference-panel harmonisation

Following GWAS quality control and reference-panel preparation, LD clumping was performed independently for each GWAS dataset using PLINK 2 and the European 1000 Genomes 30× reference panel.

The first stage involved mapping GWAS variants to the reference panel.

Variant matching required concordance of chromosome, genomic position, and allele identity in GRCh38 coordinates.

Variants without a compatible reference-panel representation were excluded from the LD-clumping calculation and recorded in a separate audit.

Unmatched variants were classified according to their mapping failure, including absence from the reference panel, incompatible alleles, or ambiguous mapping.

This harmonisation was performed separately for each GWAS study.

3.4.2 LD-clumping parameters

Clumping was performed using genome-wide significant variants as candidate index variants.

A primary significance threshold of $P_1 = 5 \times 10^{-8}$ was applied to identify eligible index variants.

A secondary threshold of $P_2 = 0.01$ was used for additional variants considered in the clumping procedure.

Pairwise LD was estimated using the ancestry-matched European reference panel. The clumping threshold was set to $r^2 = 0.1$, with a distance window of 1,000 kilobases.

For each index variant, nearby variants satisfying the secondary P-value threshold and the LD criterion were grouped within the corresponding clump.

The resulting index variants represented approximately LD-independent association signals under the specified clumping parameters.

Clumping did not establish causal variants; rather, it reduced correlated association records to representative lead variants.

Table 5 provides the parameters used for this analysis.

Table 5: Parameters used for GWAS variant matching and PLINK 2 LD clumping.

Parameter	**Setting**
Clumping software	PLINK 2
GWAS genome assembly	GRCh38
Reference panel	1000 Genomes 30×
Reference ancestry	European (EUR)
Reference sample size	503 individuals
Chromosomes	1–22
Variant mapping	Chromosome, position and allele concordance
Primary P-value threshold (P_1)	5×10^{-8}
Secondary P-value threshold (P_2)	0.01
LD threshold	$r^2 = 0.1$
Clumping window	1,000 kb
Locus-consolidation procedure	Overlapping lead-centred ±1 Mb regions merged into study-specific loci
Candidate-processing chunk size	500,000 records
Unmatched significant variants	Retained in separate mapping-audit outputs

3.4.3 Identification of genomic loci

Following LD clumping, the retained lead variants were used to define broader genomic loci.

Genomic intervals extending 1 Mb upstream and downstream of lead variants were constructed and overlapping intervals were consolidated.

Consequently, a single genomic locus could contain multiple approximately LD-independent lead variants.

A locus was therefore distinguished from an individual lead variant:

- A lead variant represented an association selected after LD clumping.
- A genomic locus represented a broader region containing one or more lead variants.

This distinction is important for subsequent fine-mapping, which evaluates candidate causal variants within genomic regions rather than assuming that the clumping lead is causal.

3.4.4 GWAS-specific clumping results

Across the 17 GWAS datasets, the QC summaries identified 13,148 genome-wide significant variant records.

Of these, 96 were reported in the three chromosome-X studies and were outside the autosomal reference-panel scope.

Among the remaining 13,052 significant autosomal records, 12,674 could be matched to the European 1000 Genomes 30× reference panel.

The clumping procedure identified a total of 59 lead variants across eight GWAS datasets with at least one retained autosomal association.

These lead variants were consolidated into 31 study-specific genomic loci.

Table 6 summarises the results for each GWAS dataset.

Table 6: LD-clumping results for each GWAS study. Representative lead positions are shown for studies with retained signals. Positions are in GRCh38 megabases (Mb), rounded to three decimal places. They are not complete locus boundaries or a complete list of all 59 lead variants.

GCST	GWS	Matched	Leads	Loci	Representative lead positions
GCST90000014	0	0	0	0	–
GCST90000015	5	4	4	4	chr6:36.725; chr15:58.516; chr18:51.140, 75.103
GCST90042668	1	1	1	1	chr17:42.146
GCST90042681	2,966	2,900	6	2	chr17:45.771, 45.904, 46.072, 46.106, 46.785
GCST90042694	0	0	0	0	–
GCST90043734	0	0	0	0	–
GCST90079817	0	0	0	0	–
GCST90104085	32	0	0	0	Chromosome X only; outside autosomal panel
GCST90104086	32	0	0	0	Chromosome X only; outside autosomal panel
GCST90104087	32	0	0	0	Chromosome X only; outside autosomal panel
GCST90281037	0	0	0	0	–
GCST90319903	3,167	3,101	14	7	chr2:168.287; chr4:89.716; chr17:45.760, 45.970, 46.072
GCST90435905	0	0	0	0	–

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GCST	GWS	Matched	Leads	Loci	Representative lead positions
GCST90473307	3,246	3,149	13	7	chr2:168.310; chr4:89.716, 89.853; chr17:45.825, 45.970
GCST90475830	3,642	3,495	13	4	chr4:89.705; chr17:45.679, 45.904, 46.053, 46.106
GCST90476961	23	22	6	4	chr2:61.028, 61.908, 62.073, 168.287; chr4:89.705
GCST90476962	2	2	2	2	chr4:89.705; chr13:36.533
Total	**13,148**	**12,674**	**59**	**31**	**Study-specific counts**

GWS denotes genome-wide significant variant records. The matched column reports significant variants successfully mapped to the autosomal LD reference. The lead and locus columns represent the respective study-specific outputs of the clumping pipeline.

The representative coordinates are examples printed by the Step05 clumping audit. Complete lead-variant identifiers and locus intervals should be reported from the corresponding per-study `lead_variants.tsv` and downstream locus definition files.

3.4.5 Comparison between GWAS significance and independent association signals

The reduction in the number of variant associations after LD clumping was substantial.

For example, GCST90475830 contained 3,642 genome-wide significant variants, of which 3,495 were successfully matched to the LD reference panel.

These mapped associations produced 13 lead variants, subsequently consolidated into four broader genomic loci.

Similarly, GCST90319903 contained 3,167 significant variants, of which 3,101 matched the reference panel. These were represented by 14 lead variants grouped into seven loci.

In the whole-genome sequencing dataset GCST90473307, 3,149 of 3,246 significant variants matched the reference, resulting in 13 lead variants distributed across seven loci.

The large difference between the number of significant variants and lead variants illustrates the contribution of LD to GWAS association patterns.

However, the 59 identified leads should not be interpreted as 59 independent biological discoveries. Their independence was defined by the selected LD threshold and reference panel, and some loci overlap between different GWAS datasets.

3.4.6 Unmapped association signals

Although most genome-wide significant autosomal variants matched the reference, a subset could not be mapped because of missing reference variants, incompatible alleles, or ambiguous representations.

In most instances, unmatched significant variants occurred close to retained lead variants and therefore remained represented at the regional level.

However, three significant association regions were flagged as lacking a mapped lead within the 1 Mb search window:

- GCST90000015: chr2:199,727,717, with an association P-value of approximately 2.4×10^{-8}.

- GCST90473307: chr3:128,656,141, with an association P-value of approximately 5.7×10^{-9}.
- GCST90473307: chr12:37,686,151, with an association P-value of approximately 7.1×10^{-9}.

These regions were not represented among the retained clumping leads because their significant variants could not be matched to the available reference panel.

Their absence from the clumping output does not invalidate the original GWAS associations. Rather, it identifies a limitation of reference-panel coverage that may require further investigation.

3.4.7 Output for subsequent fine-mapping

The output of this stage consisted of study-specific lead-variant tables, reference-mapped association variants, genomic locus definitions, and audit reports describing unmatched significant variants.

These outputs were used to establish the genomic regions subsequently considered for statistical fine-mapping.

Whereas LD clumping identified representative association signals, fine-mapping was intended to assign posterior support to candidate causal variants within those regions.

Thus, the principal outcome of the initial GWAS processing stages was a harmonised collection of study-specific genomic association regions and their supporting variant-level statistics, rather than a definitive set of causal genes or variants.

3.5 Statistical fine-mapping of GWAS loci (workflow step 6)

3.5.1 Rationale for statistical fine-mapping

Following linkage disequilibrium (LD) clumping, the identified genomic regions were subjected to statistical fine-mapping to prioritise variants potentially responsible for the observed GWAS associations.

Although LD clumping identifies representative lead variants, these variants are not necessarily causal. Several variants in the same genomic region may display similar association statistics because of correlations between their alleles.

Fine-mapping therefore seeks to distinguish candidate causal variants from correlated variants by jointly modelling GWAS association statistics and local LD structure.

We employed Sum of Single Effects Regression using Summary Statistics (SuSiE-RSS), implemented in the `susieR` R package [1, 2].

SuSiE-RSS permits multiple association signals within a locus and estimates the posterior inclusion probability (PIP) of each candidate variant, together with credible sets representing individual association signals.

3.5.2 Fine-mapping workflow

Fine-mapping was performed independently for each eligible GWAS dataset using the harmonised GRCh38 summary statistics and the European 1000 Genomes high-coverage (30×) reference panel.

First, genomic loci were reconstructed from the Step 4 lead variants. Each lead variant was extended by 1 Mb upstream and downstream, and overlapping intervals were merged into study-specific loci.

All available quality-controlled GWAS variants within each locus were extracted, including variants that did not individually achieve genome-wide significance.

Variants were subsequently matched to the LD reference panel by genomic position and allele identity. Association Z-scores were oriented consistently with the reference-panel allele coding.

Reference variants were filtered using a minor allele frequency threshold of 0.01 and a maximum genotype missingness rate of 0.05. A maximum of 15,000 variants per locus was imposed for LD matrix construction.

For loci exceeding this limit, a deterministic selection procedure retained variants surrounding the independent lead positions. Such downsampling was recorded and flagged during quality assessment.

Signed pairwise LD correlations were calculated using PLINK 2 and supplied to SuSiE-RSS together with the harmonised GWAS Z-scores and study-specific sample size.

For binary traits, effective sample size was used where available:

$$N_{\text{eff}} = \frac{4N_{\text{cases}}N_{\text{controls}}}{N_{\text{cases}} + N_{\text{controls}}}. \tag{2}$$

For continuous traits, the total sample size was used.

SuSiE-RSS was executed using a target credible-set coverage of 95%, with an initial allowance of ten single-effect components, increased according to the number of clumped leads where necessary, up to a maximum of twenty components.

Residual variance estimation was disabled because the LD matrices were obtained from an external reference panel.

The posterior inclusion probability for variant j can be expressed conceptually as

$$\text{PIP}_j = \Pr\left(\beta_j \neq 0 \mid \mathbf{z}, \mathbf{R}, N\right), \tag{3}$$

where $\mathbf{z}$ denotes the vector of GWAS association Z-scores, $\mathbf{R}$ represents the signed LD correlation matrix, and N denotes the effective or total GWAS sample size, as appropriate.

For each inferred effect, SuSiE-RSS was used to construct a credible set with a nominal posterior coverage of 95%.

Model convergence, LD consistency, possible allele orientation errors, credible-set purity, and retention of the original clumping leads were examined as quality-control criteria.

3.5.3 GWAS-specific fine-mapping results

Across the 17 original GWAS datasets, eight contained autosomal association loci eligible for fine-mapping.

The 59 lead variants identified by LD clumping were represented by 32 study-specific genomic loci in the current fine-mapping run. The preceding clumping audit recorded 31 study-specific loci. In particular, GCST90473307 had seven loci in the clumping summary but eight in the fine-mapping inventory. The precise origin of the extra reconstructed interval has not been

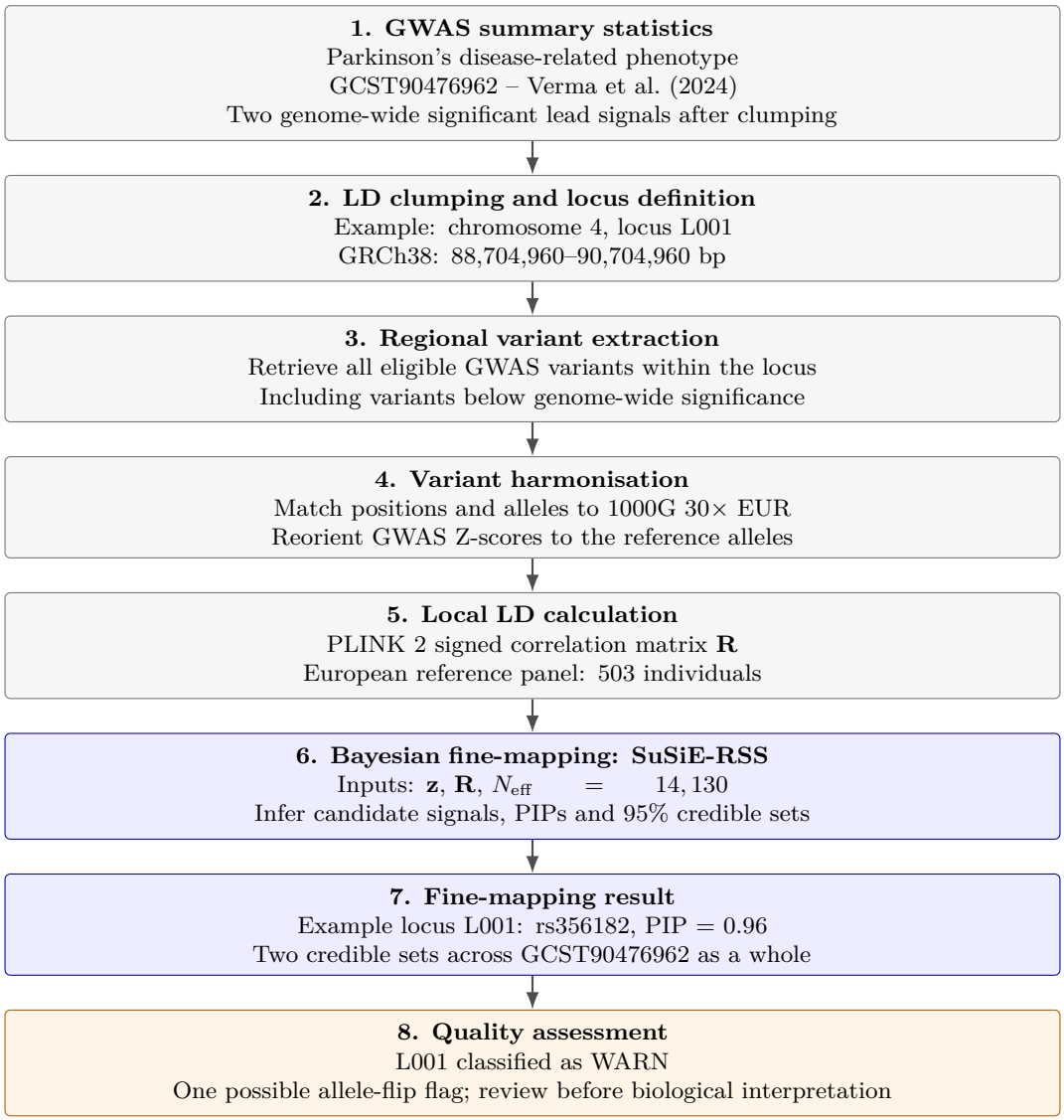


Figure 2: GWAS2Mech statistical fine-mapping workflow illustrated using the observed results for GCST90476962 (Verma et al., 2024). The original association region on chromosome 4 was analysed using SuSiE-RSS with European reference LD. The variant rs356182 achieved a posterior inclusion probability of approximately 0.96. The reported credible-set count refers to the entire GWAS dataset, not exclusively to the illustrated locus. A QC warning indicates that the locus requires further assessment before causal interpretation.

reconciled against the source locus-definition files; these are stage-specific observed counts and must not be portrayed as a single internally identical set of intervals.

All 32 loci completed SuSiE-RSS processing without a recorded locus-level execution failure.

Fine-mapping identified a total of 69 study-specific 95% credible sets.

Across the fine-mapped study records, 60 variants had PIP greater than 0.5, of which 52 had PIP greater than 0.9.

These totals represent study-specific variant records and credible sets rather than unique variants or independent biological signals across all studies.

Table 7 summarises the study-specific results.

Table 7: Statistical fine-mapping results across eight Parkinson's disease and related GWAS datasets. GWAS accession labels are the primary provenance; original-publication attributions for all accession rows should be checked against the Catalog metadata before external submission. Leads denote variants retained after LD clumping; loci denote study-specific genomic regions; CS denotes 95% credible sets. High-PIP variants are counted within each GWAS. QC is reported as PASS/WARN locus counts.

GWAS accession	Publication	Leads	Loci	N used	CS	PIP > 0.9	PASS	WARN
GCST90000015	Alfradique-Dunham (2021) [10]	4	4	3,212	0	0	0	4
GCST90042668	Jiang (2021) [11]	1	1	38,832	0	0	0	1
GCST90042681	Jiang (2021) [11]	6	2	27,054	8	6	0	2
GCST90319903	Doostparast Torshizi (2024) [11]	14	7	11,253	19	16	0	7
GCST90473307	UKB WGS Consortium (2025) [12]	13	8	16,713	15	11	0	8
GCST90475830	Verma (2024) [13]	13	4	34,141	14	11	1	3
GCST90476961	Verma (2024) [13]	6	4	15,980	11	7	1	3
GCST90476962	Verma (2024) [13]	2	2	14,130	2	1	0	2
Total	–	**59**	**32**	–	**69**	**52**	**2**	**30**

The sample sizes shown in Table 7 are the values supplied to SuSiE-RSS, rather than necessarily the total number of participants in each GWAS. For binary studies these generally correspond to the effective sample size, whereas the continuous motor-subtype study used the total sample size.

The number of credible sets varied substantially between studies.

GCST90319903 produced 19 credible sets across seven fine-mapped loci, while GCST90473307 produced 15 credible sets across eight loci.

GCST90475830 yielded 14 credible sets across four loci, and GCST90476961 yielded 11 credible sets across four loci.

In contrast, GCST90000015 and GCST90042668 produced no retained credible sets despite completing fine-mapping.

Thus, the existence of a genome-wide significant lead association did not guarantee the identification of a sufficiently resolved credible set.

3.5.4 Genomic distribution of fine-mapped loci

The current fine-mapping analysis defined 32 study-specific regions across the eight eligible GWAS datasets.

Table 8 reports the locus identifiers, genomic intervals and fine-mapping quality-control classifications.

Coordinates are reported in GRCh38 base pairs. Because fine-mapping was performed independently for each GWAS dataset, overlapping intervals in different studies were retained as separate study-specific loci.

Table 8: Genomic intervals of the 32 study-specific loci processed using SuSiE-RSS; the upstream clumping audit reported 31 loci (one additional interval in GCST90473307 at this stage remains to be reconciled). Coordinates are based on GRCh38. The two intervals absent from the supplied summary log require extraction from their loci definition files.

GWAS accession	Locus	Genomic region (GRCh38, bp)	QC
GCST90000015	L001	chr6:35,725,362–37,725,362	WARN
GCST90000015	L002	chr15:57,515,688–59,515,688	WARN
GCST90000015	L003	chr18:50,139,811–52,139,811	WARN
GCST90000015	L004	chr18:74,103,246–76,103,246	WARN
GCST90042668	L001	chr17:41,146,497–43,146,497	WARN
GCST90042681	L001	chr12:39,340,400–41,340,400	WARN
GCST90042681	L002	chr17:44,771,384–47,785,247	WARN
GCST90319903	L001	chr1:154,236,376–156,236,376	WARN
GCST90319903	L002	chr2:167,286,525–169,286,525	WARN
GCST90319903	L003	chr4:88,716,450–90,821,664	WARN
GCST90319903	L004	chr12:39,340,400–41,340,400	WARN
GCST90319903	L005	chr14:53,881,109–55,881,109	WARN
GCST90319903	L006	chr17:44,759,587–47,785,767	WARN
GCST90319903	L007	chr19:43,908,684–45,908,684	WARN
GCST90473307	L001	chr1:154,236,376–156,236,376	WARN
GCST90473307	L002	chr2:167,309,772–169,309,772	WARN
GCST90473307	L003	chr3:164,216,729–166,216,729	WARN
GCST90473307	L004	chr4:1–1,912,343	WARN
GCST90473307	L005	chr4:88,716,450–90,852,931	WARN
GCST90473307	L006	chr12:39,340,400–41,340,400	WARN
GCST90473307	L007	chr14:53,870,033–55,870,033	WARN
GCST90473307	L008	chr17:44,486,240–46,969,671	WARN
GCST90475830	L001	chr2:167,286,525–169,286,525	WARN
GCST90475830	L002	chr4:88,704,947–90,704,947	PASS
GCST90475830	L003	chr12:39,340,400–41,340,400	WARN
GCST90475830	L004	chr17:44,604,405–47,709,947	WARN
GCST90476961	L001	chr2:60,028,470–63,072,723	WARN
GCST90476961	L002	chr2:167,286,525–169,286,525	WARN
GCST90476961	L003	chr4:88,704,960–90,704,960	PASS
GCST90476961	L004	chr17:44,625,426–46,625,426	WARN
GCST90476962	L001	chr4:88,704,960–90,704,960	WARN
GCST90476962	L002	chr13:35,532,864–37,532,864	WARN

3.5.5 Posterior prioritisation of candidate variants

Fine-mapping produced study-specific rankings of candidate variants according to their posterior inclusion probabilities.

Several variants achieved very high PIPs in the Parkinson's disease-associated regions.

For example, in GCST90476962, rs356182 was the highest-ranked reported variant within locus L001 on chromosome 4, achieving a PIP of approximately 0.96.

Within GCST90319903, variants including rs374035281, rs16940802 and rs71363566 were reported among the highest-PIP candidates.

Likewise, GCST90475830 and GCST90476961 reported high-PIP variants including rs62058964 and rs66781068.

However, these probabilities are conditional on the fine-mapping model, variant inclusion, LD reference and quality of the association statistics.

A high PIP must therefore not be interpreted as experimental proof of a causal variant.

Furthermore, variant-level support cannot be assumed to represent independent replication when different GWAS analyses include overlapping participants.

3.5.6 Fine-mapping quality and limitations

Although all 32 loci completed computational processing, only two satisfied all implemented fine-mapping quality-control criteria.

The remaining 30 loci were classified as WARN.

Five locus-level SuSiE-RSS fits failed to converge within the configured maximum of 500 iterations.

Across the study-specific analyses, 33 possible allele-flip events were flagged by the diagnostic procedures. These represent potential inconsistencies rather than confirmed allele-coding errors.

Warnings also included discrepancies between GWAS association Z-scores and reference-panel LD, missing original lead variants after reference matching, and high-PIP variants outside retained credible sets.

For example, GCST90473307 locus L008, located at chr17:44,486,240–46,969,671, did not converge within 500 iterations and was flagged for three possible allele flips and an original lead position missing from the final LD variant set.

Similarly, GCST90319903 locus L006, located at chr17:44,759,587–47,785,767, did not converge and was flagged for a possible allele flip and missing clumping lead positions.

These findings demonstrate that numerical completion of a fine-mapping run does not necessarily imply reliable resolution of its causal signals.

The posterior probabilities and credible sets should therefore be interpreted in conjunction with their locus-specific quality-control results.

In particular, high-confidence causal claims should not be based solely on PIP thresholds when the corresponding loci show non-convergence, possible allele mismatches, or inconsistencies between summary statistics and reference LD.

3.5.7 Worked example of statistical fine-mapping at chromosome 4

To illustrate the distinction between lead-variant selection and fine-mapping, we examined the overlapping chromosome 4 regions in GCST90475830, GCST90476961, and GCST90476962 [13]. The lead variants reported by GCST90475830 (locus L002, `4:89704947:G:C`) and GCST90476961 (locus L003, `4:89704960:G:A`) were located only 13 base pairs apart in GRCh38 coordinates. These are different positions, not an identical lead variant. Their lead-centred 2-Mb intervals

overlap extensively. The GCST90476962 L001 interval spanned chr4:88,704,960–90,704,960 and was analysed separately using its own GWAS summary statistics.

In each region, all available quality-controlled regional variants, not only the lead SNP, were matched to the European-ancestry 1000 Genomes 30× reference panel. An LD correlation matrix $\mathbf{R}$ was computed for the harmonised variants; the corresponding association Z-scores $\mathbf{z}$ and study-specific sample size N were supplied to SuSiE-RSS [2]. For GCST90476962, the recorded effective sample size was

$$N_{\text{eff}} = \frac{4(3,573)(312,095)}{3,573 + 312,095} \approx 14,130. \tag{4}$$

In the Sum of Single Effects model, a regression effect is represented as a sum of L single-effect components:

$$\boldsymbol{\beta} = \sum_{\ell=1}^{L} \boldsymbol{\beta}_\ell, \qquad \|\boldsymbol{\beta}_\ell\|_0 \leq 1. \tag{5}$$

Let $\alpha_{\ell j}$ denote the posterior probability that variant j is selected for component ℓ. The posterior inclusion probability (PIP) is then

$$\text{PIP}_j = 1 - \prod_{\ell=1}^{L}(1 - \alpha_{\ell j}), \tag{6}$$

whereas a nominal 95% credible set C_ℓ collects variants whose component-specific posterior probabilities satisfy

$$\sum_{j \in C_\ell} \alpha_{\ell j} \geq 0.95. \tag{7}$$

Thus, the output is a posterior prioritisation of variants and credible sets, rather than a second genome-wide significance test.

The observed Step06 output for GCST90476962 L001 ranked `rs356182` first (PIP approximately 0.96); the subsequent reported candidates `rs920624`, `rs34991691`, `rs6828271`, and `rs6532191` had rounded PIPs of 0.04, 0.04, 0.04, and 0.03, respectively. Because the locus was labelled WARN (one possible allele flip), the fine-mapping result requires quality review before strong causal interpretation.

3.5.8 Recurrence of overlapping loci across GWAS datasets

To summarise cross-study recurrence, we merged overlapping study-specific Step06 locus intervals on the same GRCh38 chromosome. For each resulting physical region r, the recurrence frequency was calculated as the number of *distinct GWAS Catalog accessions* whose fine-mapped locus intervals overlapped the region:

$$F_r = \left|\{s : \exists\, L_{sk} \text{ such that } L_{sk} \cap r \neq \varnothing\}\right|. \tag{8}$$

Each accession contributed at most once to a merged physical region. This is an overlap-based recurrence measure, not a meta-analysis, an exact-SNP replication count, or evidence of independent cohorts.

The 32 study-specific fine-mapping intervals collapsed into 16 physical regions: 10 observed

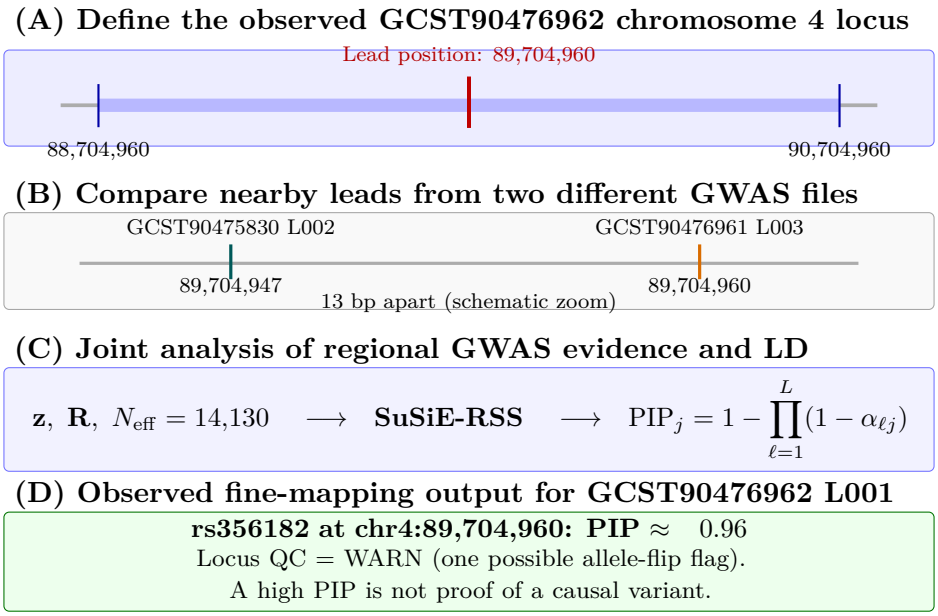


Figure 3: Worked example of the implemented GWAS2Mech fine-mapping procedure, using observed GRCh38 coordinates and Step06 results. Only panel B uses a non-scaled close-up to distinguish the two leads 13 bp apart. Panel D uses the reported high-PIP candidate from GCST90476962 L001; it is not a P-value and does not constitute proof of causality.

in one GWAS, two in two GWAS, two in four GWAS, and two in five GWAS. The most recurrent regions were on chromosomes 4 and 17 (five GWAS each), followed by chromosomes 2 and 12 (four GWAS each). Table 9 reports all 16 regions.

Table 9: Recurrence of overlapping GRCh38 physical regions across the eight fine-mapped GWAS accessions. Interval boundaries are the union of the observed, overlapping Step06 study-specific loci. Frequency (F) counts distinct GWAS files, not independent samples.

Region	Merged interval (GRCh38)	Supporting GWAS accessions	F
R01	chr1:154,236,376–156,236,376	GCST90319903, GCST90473307	2
R02	chr2:60,028,470–63,072,723	GCST90476961	1
R03	**chr2:167,286,525–169,309,772**	GCST90319903, GCST90473307, GCST90475830, GCST90476961	**4**
R04	chr3:164,216,729–166,216,729	GCST90473307	1
R05	chr4:1–1,912,343	GCST90473307	1
R06	**chr4:88,704,947–90,852,931**	GCST90319903, GCST90473307, GCST90475830, GCST90476961, GCST90476962	**5**
R07	chr6:35,725,362–37,725,362	GCST90000015	1
R08	**chr12:39,340,400–41,340,400**	GCST90042681, GCST90319903, GCST90473307, GCST90475830	**4**
R09	chr13:35,532,864–37,532,864	GCST90476962	1
R10	chr14:53,870,033–55,881,109	GCST90319903, GCST90473307	2
R11	chr15:57,515,688–59,515,688	GCST90000015	1
R12	chr17:41,146,497–43,146,497	GCST90042668	1
R13	**chr17:44,486,240–47,785,767**	GCST90042681, GCST90319903, GCST90473307, GCST90475830, GCST90476961	**5**
R14	chr18:50,139,811–52,139,811	GCST90000015	1
R15	chr18:74,103,246–76,103,246	GCST90000015	1

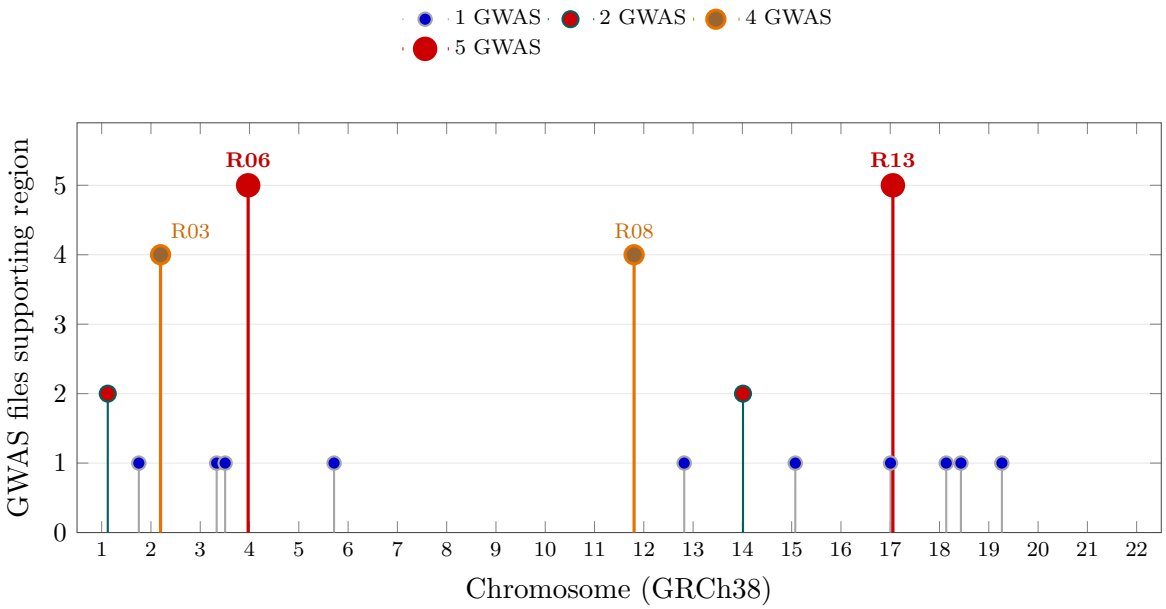


Figure 4: Manhattan-style genome-wide recurrence map of the 16 cross-study merged physical regions. The horizontal axis positions each regional midpoint proportionally within its GRCh38 chromosome; chromosomes occupy equal display widths. The vertical axis counts GWAS accessions containing an overlapping fine-mapped locus, not $-\log_{10}(P)$. Red indicates the two most recurrent regions (R06, chr4; R13, chr17; five GWAS each), and orange marks R03 (chr2) and R08 (chr12), each observed in four GWAS.

These recurrent regions indicate that several related GWAS analyses implicate overlapping genomic intervals. They do not establish that the same lead allele or causal variant was selected in every study, nor do they provide independent replication because some GWAS use related phenotypes and potentially overlapping cohorts. The posterior prioritisation and QC of the study-specific fine-mapping results should therefore be evaluated separately from regional recurrence frequency.

3.5.9 Output for downstream molecular interpretation

The fine-mapping stage generated variant-level posterior probabilities, credible-set memberships, locus definitions and quality-control diagnostics.

These outputs provide the variant-level statistical evidence required for the subsequent stages of GWAS2Mech, including functional variant annotation, molecular QTL colocalisation and prioritisation of candidate disease-associated genes.

Fine-mapping alone does not identify the molecular mechanism of disease association.

Instead, the prioritised variants are carried forward for integration with regulatory annotations, transcriptomic and other molecular QTL evidence, and additional functional genomic resources.

3.6 Functional annotation of fine-mapped variants (workflow step 8)

3.6.1 Functional annotation resources and reference datasets

All variants retained for fine-mapping were functionally annotated at the variant level against local, GRCh38-compatible resources (Table 10). The use of `scope=all` meant that annotation

was not restricted to the original lead variants or to variants in 95% credible sets. The final column reports annotation availability or a positive interval overlap in the 448 study-specific credible-set variant records; it is not the number of independently validated functional variants.

Table 10: Functional annotation resources integrated in GWAS2Mech Step 08. All resources were used for variant-level or gene-level annotations within the eight fine-mapped GWAS datasets. The coverage column refers to the 448 study-specific 95% credible-set variant records. Positive interval overlaps and database matches are distinguished from nonmissing numerical scores.

Resource (release)	**Information retrieved**	**Example output / interpretation**	**Observed credible-set coverage**
Ensembl VEP (release 116; GRCh38) [14]	Transcript-level consequences, representative consequence, gene assignment, coding and splicing context, HGVS descriptions, SIFT/PolyPhen predictions when applicable.	Consequence term; functional category; gene symbol; HGVSc/HGVSp.	Consequence: 404/448; gene: 214/448
CADD (v1.7; SNVs and indels) [15]	Sequence-variant deleteriousness predictions using locally indexed SNV and indel reference data.	Raw and PHRED scores; high-score flag at PHRED ≥ 20.	Score: 448/448
AlphaMissense (hg38 precomputed) [16]	Predicted pathogenicity score and class for eligible missense substitutions; not a genome-wide consequence score.	Pathogenicity score; likely benign, ambiguous, or likely pathogenic class.	Score: 2/448
ClinVar (local GRCh38 VCF) [17]	Allele-matched clinical significance, reported conditions, and review information from the locally stored release.	Clinical significance; pathogenic/likely pathogenic indicator.	Database records: 17/448
ENCODE SCREEN (cCRE V4) [18]	Overlap with candidate cis-regulatory elements, including promoter-like and enhancer-like elements and accessible/CTCF-associated elements.	Candidate cis-regulatory element (cCRE) class; PLS, pELS, dELS, CA, etc.	Overlapping: 99/448
UCSC phyloP (100-way; hg38)	Base-wise evolutionary conservation inferred from vertebrate multiple-sequence alignments.	phyloP score; conservation flag at score ≥ 2.	Score: 446/448
gnomAD population frequencies (via VEP)	Variant population allele frequencies obtained from the available VEP-linked population-frequency annotations, including non-Finnish European frequency when present.	Overall and ancestry-specific allele-frequency fields.	Frequency: 419/448
gnomAD gene constraint (v4.1)	Gene-level selective constraint attached to VEP-mapped genes, including loss-of-function intolerance and missense constraint.	LOEUF, pLI, missense Z-score.	Constraint: 206/448
RegulomeDB (v2.2; local GRCh38)	Regulatory ranks, probabilities, and evidence for matching alleles from the local, dbSNP153 common-SNV resource; rare variants and some indels may have no entry.	Rank 1a–7; probability; strong rank 1a–1f or 2a–2f.	Rank: 417/448; strong: 229/448

The high-level functional classifications were produced by VEP, whereas CADD, AlphaMis-

sense, ClinVar, ENCODE cCREs, phyloP, gnomAD, and RegulomeDB supplied complementary predictive, clinical, evolutionary, population, and regulatory annotations. Of the 448 credible-set records, 229 (51.12%) carried a pipeline-defined strong RegulomeDB rank, and 99 (22.10%) overlapped an ENCODE cCRE. AlphaMissense covered two missense records and should not be interpreted as providing a negative prediction for the remaining non-missense records. The 1000 Genomes 30× European panel was the LD reference used during Step 06 fine-mapping and was *not* an additional functional annotation database in Step 08.

3.6.2 Variant-level annotation strategy

To connect statistical fine-mapping with potential molecular mechanisms, we functionally annotated the variants retained within the SuSiE-RSS fine-mapping regions. Annotation was performed at the *individual variant* level, rather than solely for the LD-clumping lead variants or the genomic loci. For each of the eight GWAS datasets with eligible loci, Step 08 used the study-specific Step 06 fine-mapping table with `scope=all`, retaining the harmonised chromosome, GRCh38 position, reference and alternative alleles, locus identifier, posterior inclusion probability (PIP), and 95% credible-set membership.

Annotations were first retrieved by genomic variant key (chromosome, position, reference allele and alternative allele), and then joined to the fine-mapping records for each study. Thus, the same physical variant could appear in multiple GWAS analyses with identical reference annotations but different association statistics, PIPs, and credible-set memberships. The European 1000 Genomes high-coverage 30× reference panel supplied the LD information at the *fine-mapping* stage; the functional annotations were obtained separately from local GRCh38 reference resources.

Predicted molecular consequences were obtained using Ensembl Variant Effect Predictor (VEP) release 116 with its offline GRCh38 cache [14]. The pipeline extracted transcript-level consequences, the selected representative consequence, gene identifier and symbol, transcript information, and HGVS descriptions when available. Functional categories were assigned using the pipeline's consequence classification: `CODING`, `SPLICING`, `UTR`, `INTRONIC`, `NONCODING_RNA`, `UPSTREAM`, `DOWNSTREAM`, and `OTHER`. These categories were mutually exclusive for the summary tables, whereas the underlying VEP consequence terms could overlap across transcripts.

Additional annotations were incorporated from CADD v1.7 (SNV and indel scores) [15], AlphaMissense missense predictions [16], ClinVar clinical classifications [17], ENCODE SCREEN version 4 candidate cis-regulatory elements (cCREs) [18], UCSC phyloP 100-way conservation scores, gnomAD allele frequencies and gene-level constraint metrics, and local RegulomeDB version 2.2 regulatory evidence. The annotation output retained source-specific scores and classifications; no composite functional causality score was inferred.

3.6.3 Functional evidence definitions and denominators

Protein-altering variants were identified from the VEP consequence categories, including missense and predicted loss-of-function (LoF) consequences. The prespecified annotation flags included CADD PHRED ≥ 20, AlphaMissense `likely_pathogenic`, ClinVar pathogenic/likely pathogenic (excluding conflicting classifications according to the implemented rule), phyloP 100-way ≥ 2,

overlap with an ENCODE cCRE, and RegulomeDB rankings classified by the pipeline as strong (rank classes 1a–1f or 2a–2f). These are source-specific annotation indicators and should not be interpreted as independent experimental validation of molecular activity.

The main functional interpretation focused on the variants retained within the nominal 95% SuSiE-RSS credible sets, while the full annotated variant collection was retained for comparison and downstream analyses. For functional category k, the reported credible-set proportion was

$$f_k = \frac{n_{k,\mathrm{CS}}}{N_{\mathrm{CS}}}, \qquad N_{\mathrm{CS}} = 448, \tag{9}$$

where $n_{k,\mathrm{CS}}$ counts study-specific credible-set variant records assigned to category k. The 448 records are not 448 independent causal variants: recurrent variants can appear in multiple studies, and several variants may belong to the same credible set. The denominator for annotation-source coverage similarly refers to records with a nonmissing field or a positive interval overlap, not the total number of possible predictions for the variant's biological class.

3.6.4 Annotation yield across GWAS datasets

Of the 17 original GWAS datasets, eight had fine-mapped loci and completed Step 08 functional annotation; the remaining nine had no eligible fine-mapped loci and were skipped. Across the eight datasets, 204,683 study-specific fine-mapped variant records were annotated in 32 study-specific loci. These records corresponded to 103,881 distinct genomic variant keys across datasets.

A total of 69 nominal 95% credible sets were reported by fine-mapping. Their retained variant annotations comprised 448 study-specific records representing 219 distinct genomic variant keys. At least one credible-set variant was present in 19 of the 32 loci; 13 loci had no retained credible-set variants. Across the 219 distinct credible-set variant keys, 136 were represented in two or more GWAS datasets. Recurrence indicates overlap among the analysed GWAS results and must not automatically be interpreted as replication in independent samples.

Table 11 reports the annotation yield and selected functional evidence separately for each dataset.

Table 11: Step 08 study-level annotation results. All counts refer to study-specific variant records. A 95% credible-set (CS) variant is a retained variant record flagged by SuSiE-RSS. The PIP threshold is ≥ 0.9. cCRE denotes overlap with an ENCODE candidate cis-regulatory element; RDB denotes the pipeline-defined RegulomeDB 1–2 evidence flag.

GWAS accession	Loci	All variants	CS variants	PIP ≥ 0.9	CS coding	CS cCRE	CS RDB
GCST90000015	4	23,861	0	0	0	0	0
GCST90042668	1	4,267	0	0	0	0	0
GCST90042681	2	14,601	13	6	0	4	9
GCST90319903	7	43,535	75	16	1	20	41
GCST90473307	8	49,610	89	11	1	17	43
GCST90475830	4	28,943	28	11	0	10	15
GCST90476961	4	26,557	129	7	0	27	66
GCST90476962	2	13,309	114	1	0	21	55
Total	**32**	**204,683**	**448**	**52**	**2**	**99**	**229**

3.6.5 Predominance of noncoding functional consequences

Among the 448 credible-set variant records, 197 (43.97%) were assigned to `NONCODING_RNA` and 170 (37.95%) to `INTRONIC` categories, collectively accounting for 367 records (81.92%). The remaining categories comprised 44 `OTHER` (9.82%), 22 `UPSTREAM` (4.91%), seven `UTR` (1.56%), six `DOWNSTREAM` (1.34%), and two `CODING` (0.45%) records. No credible-set record was assigned to the pipeline's `SPLICING` category.

The exclusive category `NONCODING_RNA` should not be interpreted as disjoint from an intronic *VEP consequence*: for example, a variant can be intronic in a noncoding transcript and be classified as `NONCODING_RNA` by the pipeline. At the level of potentially overlapping VEP terms, `intron_variant` was recorded 360 times (80.36% of 448 records) and a noncoding-transcript VEP consequence 190 times (42.41%). Because VEP terms can co-occur, the percentages of these detailed terms need not sum to 100%.

Table 12 reports the functional classification for each study, and Table 13 provides the overall comparison between the complete annotated variant collection and the credible-set subset.

Table 12: Functional categories of 95% credible-set variant records within each GWAS. Categories are mutually exclusive by the Step 08 classification procedure; zero denotes no record assigned to that category. NC RNA, noncoding RNA.

GWAS	NC RNA	Intronic	Other	Upstream	Downstream	UTR	Coding	Total
GCST90000015	0	0	0	0	0	0	0	0
GCST90042668	0	0	0	0	0	0	0	0
GCST90042681	3	4	5	0	0	1	0	13
GCST90319903	13	42	10	7	1	1	1	75
GCST90473307	43	33	7	3	1	1	1	89
GCST90475830	9	6	9	0	3	1	0	28
GCST90476961	67	40	13	6	1	2	0	129
GCST90476962	62	45	0	6	0	1	0	114
Total	**197**	**170**	**44**	**22**	**6**	**7**	**2**	**448**

Table 13: Functional consequence composition in the full annotated collection and in the 95% credible-set subset. All values count study-specific records. The percentage column uses the 448 credible-set records as denominator.

Functional class	All annotated	CS variants	CS (%)
Noncoding RNA	46,137	197	43.97
Intronic	108,582	170	37.95
Other/unassigned	27,386	44	9.82
Upstream	9,197	22	4.91
Untranslated region	3,189	7	1.56
Downstream	7,316	6	1.34
Coding	2,386	2	0.45
Splicing	490	0	0.00
Total	**204,683**	**448**	**100.00**

3.6.6 Regulatory, deleteriousness, and clinical annotations

Of the 448 credible-set variant records, 99 (22.10%) overlapped ENCODE cCREs, while 229 (51.12%) met the pipeline-defined strong RegulomeDB criterion. Five records (1.12%) met the phyloP conservation threshold of at least 2. In contrast, no credible-set record reached CADD PHRED ≥ 20, was classified as AlphaMissense likely pathogenic, or carried an unconflicted pathogenic/likely pathogenic ClinVar flag under the implemented annotation rules. These features are not exclusive: a single variant may overlap a cCRE and also have a RegulomeDB flag.

The restriction to credible sets substantially changed these functional counts relative to all fine-mapped variant records. For example, 652 records in the full annotation collection had CADD PHRED ≥ 20 and 84 had a pipeline-defined LoF consequence, but none of these records belonged to the retained 95% credible-set variant subset. The lack of such evidence within the CS subset is not evidence that the associated regions lack biological function.

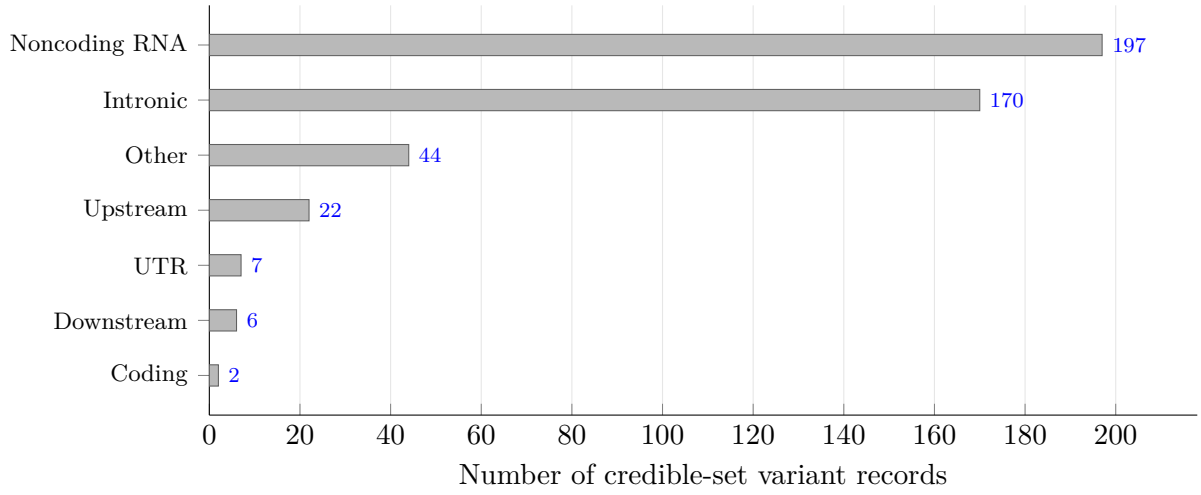


Figure 5: Mutually exclusive Step 08 functional categories among the 448 study-specific 95% credible-set variant records. The noncoding RNA category can nevertheless include VEP intron-variant terms on noncoding transcripts.

Table 14: Functional annotation flags in the full fine-mapped variant collection versus the 95% credible-set subset. Flags are non-exclusive. RegulomeDB 1–2 indicates the implemented ranking rule, rather than experimentally confirmed causal regulatory effects.

Annotation criterion	All records	CS records	CS (%)
Loss of function	84	0	0.00
Coding	2,470	2	0.45
Missense	1,214	2	0.45
Splicing	490	0	0.00
CADD PHRED ≥ 20	652	0	0.00
AlphaMissense likely pathogenic	26	0	0.00
ClinVar pathogenic/likely pathogenic	1	0	0.00
ENCODE cCRE overlap	51,615	99	22.10
phyloP 100-way ≥ 2	2,718	5	1.12
RegulomeDB strong (1–2)	92,025	229	51.12

3.6.7 Annotation completeness and resource coverage

The completeness of annotation differed by resource and type of functional effect. CADD PHRED was available for all 448 credible-set records, phyloP scores for 446 (99.55%), RegulomeDB rankings for 417 (93.08%), and gnomAD population allele frequencies for 419 (93.53%). A representative VEP consequence was available for 404 (90.18%), while a selected VEP gene symbol was available for 214 (47.77%).

AlphaMissense values were present for only two credible-set records; both represented the same missense variant in separate studies. This is expected to be a highly consequence-dependent resource, so availability relative to all 448 variants should not be used as a measure of its accuracy or completeness for eligible missense substitutions. Likewise, a missing gene symbol or empty representative VEP consequence is not evidence that a variant is biologically inert.

The local annotation provenance recorded successful resource access for CADD, AlphaMissense, ClinVar, ENCODE cCRE, phyloP, RegulomeDB, and gnomAD gene constraint across eight datasets. The VEP provenance status field was reported as `UNKNOWN` by the audit; nevertheless, VEP consequence fields were present for 404 credible-set records. The VEP status field should therefore be treated as an incomplete audit-status field rather than an indication that VEP was unavailable or failed.

Table 15: Coverage of annotation fields among all fine-mapped records and the 95% credible-set subset. Coverage reflects nonmissing values, except cCRE where it represents a positive interval overlap; consequently, the cCRE row is an overlap count rather than an assay availability estimate.

Annotation field	All n	All (%)	CS n	CS (%)
VEP representative consequence	177,297	86.6	404	90.2
VEP selected gene symbol	145,064	70.9	214	47.8
CADD PHRED score	204,679	100.0	448	100.0
AlphaMissense score	852	0.4	2	0.4
ClinVar classification	4,638	2.3	17	3.8
ENCODE cCRE overlap	51,615	25.2	99	22.1
phyloP 100-way score	203,418	99.4	446	99.6
RegulomeDB ranking	192,612	94.1	417	93.1
gnomAD allele frequency	194,339	94.9	419	93.5
gnomAD gene constraint	126,012	61.6	206	46.0

3.6.8 Locus-level annotation and candidate variants

The 32 study-specific loci contained between 3,722 and 9,332 annotated variant records each in the observed run. Table 16 reports every locus, its annotated variant count, number of variants retained in nominal 95% credible sets, highest-PIP record and representative VEP gene and consequence. The top variant is selected using PIP across *all* annotated regional variants; it is not necessarily a member of a retained credible set. Moreover, the absence of a credible set does not imply the absence of a strong marginal association.

Table 16: Complete study-specific functional annotation summary for the 32 fine-mapped loci. N is the number of variant records annotated at that locus; CS is the number of variants marked as members of 95% credible sets; $N_{0.9}$ is the number of records with PIP ≥ 0.9. The reported gene and consequence are the selected representative VEP annotation of the top-PIP variant, not an established causal target.

GWAS	Locus	N	CS	Top variant/rsID	PIP	$N_{0.9}$	Gene	VEP consequence
GCST90000015	L001	5,737	0	`6:35726373:G:A`	0.0000	0	–	downstream gene
GCST90000015	L002	6,295	0	`15:57805106:G:A`	0.0135	0	–	–
GCST90000015	L003	5,075	0	`18:51304051:C:T`	0.0142	0	–	–
GCST90000015	L004	6,754	0	`18:75158255:C:G`	0.0190	0	–	intron; noncoding transcript
GCST90042668	L001	4,267	0	`17:41147152:G:C`	0.0000	0	KRTAP4-5	downstream gene
GCST90042681	L001	6,485	0	`12:40795186:C:A`	0.0402	0	CNTN1	intron
GCST90042681	L002	8,116	13	`17:46026338:G:A`	1.0000	6	MAPT	3’ UTR
GCST90319903	L001	3,722	4	`rs2230288`	0.4927	0	GBA1	missense
GCST90319903	L002	5,999	6	`rs9917256`	1.0000	6	–	–
GCST90319903	L003	6,734	29	`rs356219`	0.9260	1	–	intron; noncoding transcript
GCST90319903	L004	6,832	0	`rs1491942`	0.0292	0	LRRK2	intron
GCST90319903	L005	5,954	26	`rs7155501`	0.2504	0	GCH1	intron
GCST90319903	L006	8,375	10	`rs16940802`	1.0000	8	MAPT	3’ UTR
GCST90319903	L007	5,919	0	`rs429358`	0.9474	1	APOE	missense
GCST90473307	L001	3,819	4	`rs2230288`	0.6455	0	GBA1	missense
GCST90473307	L002	6,251	6	`rs6722770`	0.9351	1	–	intron; noncoding transcript
GCST90473307	L003	6,112	0	`rs1086091`	0.0000	0	–	–
GCST90473307	L004	6,512	1	`rs11248057`	0.9722	1	GAK	intron
GCST90473307	L005	6,971	56	`rs356219`	0.6625	0	–	intron; noncoding transcript
GCST90473307	L006	6,849	0	`rs117463377`	0.0523	0	KIF21A	intron
GCST90473307	L007	6,223	13	`rs2183081`	0.1643	0	GCH1	intron
GCST90473307	L008	6,873	9	`rs2668679`	1.0000	9	–	intron; noncoding transcript
GCST90475830	L001	6,204	6	`rs9917256`	1.0000	6	–	–
GCST90475830	L002	6,397	13	`rs356183`	0.3641	0	–	intron; noncoding transcript
GCST90475830	L003	7,010	0	`rs1874558`	0.0511	0	SLC2A13	intron
GCST90475830	L004	9,332	9	`rs566242433`	1.0000	5	–	–
GCST90476961	L001	7,572	0	`rs139198918`	0.4801	0	–	intron; noncoding transcript
GCST90476961	L002	6,205	17	`rs9917256`	0.1591	0	–	–

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Table continued from previous page

GWAS	Locus	N	CS	Top variant/rsID	PIP	$N_{0.9}$	Gene	VEP consequence
GCST90476961	L003	6,397	103	rs356182	0.4186	0	–	intron; noncoding transcript
GCST90476961	L004	6,383	9	rs566242433	1.0000	7	–	–
GCST90476962	L001	6,398	114	rs356182	0.9565	1	–	intron; noncoding transcript
GCST90476962	L002	6,911	0	rs145240445	0.0072	0	–	–

3.6.9 Representative variants and biological interpretation

Functional annotation implicated both protein-coding and regulatory candidate mechanisms, with marked variation in statistical support across studies. At the chromosome 1 *GBA1* region, `rs2230288` (GRCh38 chr1:155,236,376:C:T) was annotated as a missense variant in both GCST90319903 (PIP 0.4927) and GCST90473307 (PIP 0.6455). Both records were included in credible sets and together accounted for the two coding/missense credible-set *records* reported in Table 14; there was one distinct missense variant key. Its CADD PHRED score was 19.46, AlphaMissense classified it as `likely_benign`, and ClinVar reported conflicting classifications, including risk-factor interpretations. These heterogeneous annotations should not be collapsed into a single pathogenicity claim.

In the chromosome 17 *MAPT* region, `rs16940802` (chr17:46,026,338:G:A) was annotated as a 3′-UTR variant with PIP 1.0000 in GCST90319903 and an overlapping ENCODE distal enhancer-like element (dELS) with a RegulomeDB 1f ranking. The same GRCh38 variant key was also detected in GCST90042681 with PIP 1.0000; the saved primary rsID field was missing for that study. These are study-specific statistical results, not two validated independent signals.

A chromosome 4 *GAK* intronic variant, `rs11248057`, had PIP 0.9722 in GCST90473307 and overlapped an ENCODE dELS, with a RegulomeDB 1f annotation. In the chromosome 4 region encompassing *SNCA*-related candidate signals, `rs356182` had PIPs of 0.0374, 0.4186, and 0.9565 across GCST90475830, GCST90476961, and GCST90476962, respectively. It was annotated as an intronic/noncoding-transcript variant with promoter-proximal enhancer-like (pELS) overlap and RegulomeDB 1f evidence. Differences in PIP across GWAS studies reflect differences in summary statistics and fine-mapping models rather than differences in the reference functional annotation of the variant.

The chromosome 2 `rs9917256` variant was retained in credible sets from four GWAS datasets and attained PIP 1.0000 in GCST90319903 and GCST90475830; despite its strong statistical support, the output contained no representative VEP consequence or selected gene symbol (functional class `OTHER`). Such observations illustrate why statistical fine-mapping confidence and functional annotation completeness must be evaluated separately.

Conversely, the *APOE* missense variant `rs429358` attained PIP 0.9474 in GCST90319903 locus L007, but that locus contained no retained credible-set variant records in the saved Step 08 output. It was therefore *not* included in the credible-set coding/missense totals. This discordance supports the need to review SuSiE-RSS diagnostics, LD matching, and credible-set filtering before causal prioritisation.

3.6.10 Interpretation and limitations

The functional annotation results suggest that the fine-mapped Parkinson's disease-associated regions are enriched in noncoding sequence annotations, with many credible-set variants showing regulatory evidence. However, the observed proportions describe the selected credible-set collection and are not a formal statistical enrichment against a matched genomic background. The annotations also do not establish tissue-specific enhancer activity or identify the molecular target gene of a noncoding variant without supporting molecular-QTL or functional evidence.

Fine-mapping quality-control warnings must be considered when interpreting all high-PIP

Table 17: Selected observed variants illustrating the integration of statistical fine-mapping with functional annotations. "CS" indicates whether the displayed variant record was present in the 95% credible-set variant table for that GWAS; a gene symbol is the representative annotation, not a confirmed target.

GWAS	Variant	Selected gene	PIP	CS	Annotation/evidence
GCST90319903	rs2230288	*GBA1*	0.4927	Yes	Missense; CADD 19.46; conflicting ClinVar
GCST90473307	rs2230288	*GBA1*	0.6455	Yes	Same missense variant; RegulomeDB 1f
GCST90319903	rs16940802	*MAPT*	1.0000	Yes	3′-UTR; ENCODE dELS; RegulomeDB 1f
GCST90473307	rs11248057	*GAK*	0.9722	Yes	Intronic; ENCODE dELS; RegulomeDB 1f
GCST90475830	rs356182	–	0.0374	Yes	Noncoding transcript/intronic; pELS; RegulomeDB 1f
GCST90476961	rs356182	–	0.4186	Yes	Same functional annotation; different study PIP
GCST90476962	rs356182	–	0.9565	Yes	Same functional annotation; different study PIP
GCST90319903	rs9917256	–	1.0000	Yes	Class OTHER; missing VEP consequence; RegulomeDB 1f
GCST90319903	rs429358	*APOE*	0.9474	No	Missense; cCRE pELS; outside retained credible sets

variants. In the preceding Step 06 analysis, 30 of the 32 study-specific loci carried at least one warning, and five model fits did not converge within the configured 500 iterations. Such locus-level warnings, missing reference/allele matches, incomplete representative VEP consequences, and overlapping GWAS samples limit causal interpretation. The functional annotations were therefore used for hypothesis generation and prioritisation rather than as definitive evidence of pathogenicity or a validated disease mechanism.

The variant-level functional evidence and credible-set memberships provide a basis for the next stage: colocalisation against eQTL, sQTL and other molecular QTL datasets, followed by evaluation of candidate regulatory mechanisms and gene targets.

3.7 Molecular-QTL inputs and colocalisation framework (workflow steps 9–12)

3.7.1 Variant annotation inputs to molecular-QTL integration

Methodology. Following GWAS locus identification and SuSiE-RSS fine-mapping, we performed comprehensive functional annotation to characterise the potential biological consequences of variants associated with Parkinson's disease.

Fine-mapped variants were annotated using Ensembl Variant Effect Predictor (VEP, release 116) against the GRCh38 human reference genome. Complementary resources included CADD, AlphaMissense, ClinVar, ENCODE candidate cis-regulatory elements (cCREs), phyloP, RegulomeDB, and gnomAD population-frequency and gene-constraint annotations.

For each variant, the pipeline retained its genomic position, reference and alternative alleles, GWAS study, locus identifier, posterior inclusion probability (PIP), credible-set membership, predicted functional consequence, gene annotations, and available functional evidence.

The primary variant-level prioritisation focused on variants belonging to the SuSiE-RSS 95% credible sets. However, annotation was also performed on the broader collection of fine-mapped

variants to preserve regional biological information.

Annotation results. Functional annotation was completed for 204,683 study-specific fine-mapped variant records across 32 genomic loci from eight European-ancestry Parkinson's disease GWAS datasets.

SuSiE-RSS fine-mapping had identified 69 study-specific 95% credible sets, containing 448 study-specific credible-set variant records. These corresponded to 219 distinct normalised variant keys across the GWAS datasets.

The functional classification of credible-set variants was dominated by noncoding annotations, particularly noncoding-RNA and intronic consequences.

Noncoding-RNA and intronic categories together accounted for 81.92% of the credible-set records, suggesting that many fine-mapped disease-associated variants may participate in noncoding regulatory processes.

Only two study-specific credible-set records were classified as coding. Both corresponded to the *GBA1* missense variant rs2230288 represented in separate GWAS datasets.

Among credible-set records, 99 overlapped ENCODE cCREs, whereas 229 satisfied the pipeline-defined strong RegulomeDB ranking criterion. Five records met the pipeline's conservation-evidence threshold.

These functional annotations provided candidate biological interpretations but did not independently demonstrate regulatory activity or establish variant causality.

3.7.2 Integration of fine-mapping and molecular-QTL data

Colocalisation framework. To investigate potential molecular mechanisms underlying the identified GWAS associations, we integrated GWAS fine-mapping evidence with molecular quantitative trait locus (QTL) datasets.

The analysis evaluated whether a disease-associated genetic signal and a molecular phenotype were consistent with sharing an underlying causal association signal.

Five hypotheses were considered:

$$H_0 : \text{Neither trait is associated}, \tag{10}$$

$$H_1 : \text{Only the GWAS trait is associated}, \tag{11}$$

$$H_2 : \text{Only the molecular trait is associated}, \tag{12}$$

$$H_3 : \text{Both traits have distinct causal signals}, \tag{13}$$

$$H_4 : \text{Both traits share a causal signal}. \tag{14}$$

The posterior probability of colocalisation was defined as

$$PP_{H4} = P(H_4 \mid D_{\text{GWAS}}, D_{\text{QTL}}). \tag{15}$$

A comparison was classified as having strong shared-signal support when

Table 18: Functional annotation resources incorporated into GWAS2Mech. The final column reports available annotation records among the 448 study-specific GWAS credible-set variant records. Some variants occur in multiple GWAS studies. Versions should be verified against the resource provenance records before final manuscript submission.

Resource	Version	Information extracted	Records
Ensembl VEP	116, GRCh38	Gene and transcript consequences, HGVS, functional effects	404
CADD	v1.7	Predicted deleteriousness and PHRED scores	448
AlphaMissense	hg38	Missense pathogenicity predictions	2
ClinVar	GRCh38 local release	Clinical significance and associated conditions	17
ENCODE SCREEN	cCRE v4	Promoter, enhancer and other candidate regulatory overlaps	99
phyloP	100-way	Evolutionary conservation	446
RegulomeDB	v2.2	Regulatory ranking and evidence	417
gnomAD frequencies	VEP-linked	Population allele frequencies	419
gnomAD constraint	v4.1	LOEUF, pLI and missense constraint	206

Table 19: Mutually exclusive functional classifications of the 448 study-specific credible-set variant records.

Functional category	Variants	Percentage (%)
Noncoding RNA	197	43.97
Intronic	170	37.95
Other	44	9.82
Upstream	22	4.91
Untranslated region	7	1.56
Downstream	6	1.34
Coding	2	0.45
Total	448	100.00

$$PP_{H4} \geq 0.80. \tag{16}$$

Comparisons with $0.50 \leq PP_{H4} < 0.80$ were classified as suggestive.

Sensitivity to the shared-signal prior was evaluated using the more conservative prior $p_{12} = 10^{-6}$.

Three complementary molecular-QTL analyses were performed, representing bulk-tissue, cell-type-specific, and external molecular-QTL resources.

3.8 Molecular-QTL colocalisation results and integrated evidence (workflow steps 9–12)

Following functional annotation and molecular QTL colocalisation, evidence was integrated across GWAS studies, genomic loci, fine-mapped variants, molecular traits, candidate genes, and biological contexts.

The integration preserved locus identifiers, GWAS posterior inclusion probabilities, credible-set membership, functional annotations, colocalisation posterior probabilities, tissue or cell-type

labels, and evidence provenance.

Bulk-tissue, cell-type-specific, and external molecular-QTL associations were maintained as separate evidence channels before aggregation at the locus–gene level.

The integrated analysis produced 1,601 study-specific locus–gene evidence records and identified 395 unique genes in the downstream pathway-input collection.

These quantities represent computational candidate associations and do not indicate 1,601 independent loci or 395 experimentally validated disease genes.

Because the analyses included potentially overlapping GWAS cohorts, multiple tissues, repeated molecular phenotypes, and loci with fine-mapping quality-control warnings, evidence aggregation was used for candidate prioritisation rather than causal confirmation.

Candidates were subsequently selected for comparison with independent external knowledge resources to assess concordance with existing genetic and functional evidence.

3.8.1 Bulk-tissue molecular-QTL colocalisation: Step 09

Bulk-tissue colocalisation was performed using molecular QTL data obtained through the eQTL Catalogue resource snapshot, including GTEx and additional configured datasets.

The submitted resource snapshot contained 246 QTL contexts and supported analysis of gene expression (eQTL), RNA splicing (sQTL), exon usage (exonQTL), transcript isoforms (isoQTL), and available protein QTL phenotypes.

GWAS SuSiE-RSS association components were compared with provider-derived molecular-QTL SuSiE components using the `coloc.bf_bf` framework.

This signal-level approach accommodated multiple association components within a locus.

The colocalisation analysis used regional variant-level Bayes factor information, while the retained GWAS credible sets defined the eligible GWAS association components.

Loci without retained GWAS credible sets were not eligible for this analysis.

3.8.2 Cell-type-specific molecular-QTL colocalisation: Step 10

To investigate whether disease-associated genetic signals were shared with cell-type-specific expression regulation, we applied the signal-level colocalisation framework to single-cell and cell-type eQTL resources.

The submitted snapshot contained 167 QTL contexts, including data from Jerber, OneK1K, Nathan, and Cytoimmgen studies.

GWAS and QTL fine-mapping components were compared using `coloc.bf_bf`, following the same general procedure used for bulk-tissue colocalisation.

Evidence derived from bulk tissues and individual cell types was retained separately to preserve biological context.

3.8.3 External molecular-QTL colocalisation: Step 11

Additional molecular-QTL datasets were incorporated to evaluate RNA splicing, protein abundance, histone-associated regulation, and chromatin accessibility.

The submitted Step 11 resource snapshot included eight datasets spanning sQTL, pQTL, haQTL, and caQTL modalities.

Regional GWAS and molecular-QTL association statistics were harmonised by GRCh38 genomic position and alleles.

Following the credible-set-aware pipeline update, only loci containing retained GWAS credible sets were eligible for testing. Individual QTL comparisons also required at least one allele-matched GWAS credible-set variant.

Nevertheless, all eligible harmonised variants across the tested regional association window were retained for statistical colocalisation, rather than restricting the association analysis to credible-set variants alone.

External QTL colocalisation employed the approximate Bayes factor method, `coloc.abf`.

Unlike the multiple-signal analyses in Steps 09 and 10, this approach assumes a single causal association signal per trait within the tested region.

The pipeline therefore recorded assumption warnings for regions containing multiple GWAS signals or complex linkage disequilibrium.

To assess consistency between regional colocalisation and GWAS fine-mapping, we calculated the conditional shared-causal-variant posterior mass within the GWAS credible set:

$$M_{\mathrm{CS}|H4} = \sum_{v \in \mathcal{C}_{\mathrm{GWAS}} \cap \mathcal{V}_{\mathrm{shared}}} P(v \mid H_4, D_{\mathrm{GWAS}}, D_{\mathrm{QTL}}). \tag{17}$$

Here, $\mathcal{C}_{\mathrm{GWAS}}$ represents the GWAS credible-set variants and $\mathcal{V}_{\mathrm{shared}}$ represents harmonised variants available in the colocalisation analysis.

A strong external-QTL result was further prioritised as credible-set-supported when

$$PP_{H4} \geq 0.80 \quad \text{and} \quad M_{\mathrm{CS}|H4} \geq 0.50. \tag{18}$$

The posterior-mass threshold was applied as a candidate-prioritisation criterion rather than an independent statistical significance test.

3.8.4 Molecular-QTL colocalisation results

GWAS and credible-set coverage. Of the 32 study-specific GWAS loci, 19 contained at least one retained SuSiE-RSS credible-set component and were eligible for molecular QTL colocalisation.

The remaining 13 loci were not evaluated by the credible-set-dependent analyses because no GWAS credible set had been retained.

The absence of a credible set was interpreted as a fine-mapping eligibility or uncertainty outcome, not evidence for the absence of a molecular effect.

Table 20 presents the study-specific fine-mapping and molecular-QTL colocalisation results.

Overall colocalisation evidence. Across the three molecular-QTL analysis stages, 35,466 candidate trait/context comparisons were formally tested.

Of these, 456 comparisons met the strong colocalisation threshold.

Table 20: Integration of GWAS fine-mapping, functional annotation, and molecular-QTL colocalisation across eight Parkinson's disease GWAS datasets. CS, credible set; Strong, candidate molecular trait/context comparison with $PP_{H4} \geq 0.80$. The columns count study-specific records and tests, not independent causal variants or genes.

GWAS accession	Loci	CS	CS variants	Bulk strong	Cell strong	External strong
GCST90000015	4	0	0	0	0	0
GCST90042668	1	0	0	0	0	0
GCST90042681	2	8	13	14	5	4
GCST90319903	7	19	75	125	51	5
GCST90473307	8	15	89	95	7	20
GCST90475830	4	14	28	27	2	3
GCST90476961	4	11	129	47	9	4
GCST90476962	2	2	114	33	4	1
Total	32	69	448	341	78	37

The bulk-tissue analysis contributed 341 strong results, the cell-type analysis contributed 78, and the external molecular-QTL analysis contributed 37.

The number of strong colocalisation tests should not be interpreted as the number of independent causal loci, genes, or variants.

Multiple molecular traits, tissues, cell types, and GWAS studies could contribute repeated evidence for related genomic regions.

Bulk-tissue molecular-QTL findings. Step 09 completed 31,241 formal colocalisation tests, identifying 341 strong shared-signal comparisons.

Of these strong results, 158 involved transcript isoform QTLs, 99 involved exon QTLs, 43 involved gene-expression QTLs, 40 involved splicing QTLs, and one involved a protein QTL.

A total of 128 strong comparisons remained robust under the conservative shared-signal prior.

Strong bulk-tissue evidence was observed across 15 study-specific loci.

Among the notable results were molecular-QTL signals associated with the *SNCA*, *MAPT*, *GAK*, and *GBA1*-associated genomic regions.

For example, the *SNCA* locus (GCST90473307, L005) displayed strong isoQTL colocalisation in adipose tissue ($PP_{H4} \approx 0.998$), with additional support in brain substantia nigra ($PP_{H4} \approx 0.993$).

These findings supported candidate transcript-level regulatory associations at the GWAS locus, without establishing that the observed molecular effects mediate disease risk.

Cell-type-specific molecular-QTL findings. Step 10 completed 2,762 formal colocalisation tests and identified 78 strong eQTL colocalisation comparisons.

Of these, 18 remained strong under the conservative shared-signal prior.

Strong colocalisation was observed across ten study-specific loci.

Among the strongest results, expression-QTL signals associated with *LRRC37A4P* (ENSG00000214425) in floor-plate progenitor cells showed $PP_{H4} \approx 0.99994$ in the 17q21.31-associated region.

Table 21: Summary of molecular-QTL colocalisation results across Steps 09–11.

Metric	Bulk	Cell	External
Tested	31,241	2,762	1,463
Strong	341	78	37
Prior robust	128	18	27
CS-supported strong	–	–	17

At the chromosome 4 region containing *SNCA*, an eQTL for ENSG00000145335 in $CD4^+$ memory T cells showed strong shared-signal support ($PP_{H4} \approx 0.970$) for GCST90319903 locus L003.

Additional cell-type signals were detected in $CD4^+$ and $CD8^+$ T-cell contexts.

These results indicated tissue- and cell-context-specific molecular associations, although the relevance of individual cell-type associations to neuronal disease mechanisms requires additional validation.

External molecular-QTL findings. Step 11 identified 4,692 candidate molecular-QTL comparisons.

Of these, 1,463 were formally tested after credible-set eligibility and variant-overlap requirements.

An additional 2,943 candidates were not tested because no GWAS credible-set variant was shared with the harmonised molecular-QTL variants, and 286 failed the minimum regional variant-overlap requirement.

Among the completed tests, 37 achieved strong colocalisation support: 34 sQTL, two haQTL, and one pQTL comparison.

Twenty-seven strong comparisons remained supported under the conservative prior.

Seventeen strong results also met the additional GWAS credible-set posterior-mass criterion. These comprised 15 sQTL and two haQTL comparisons.

Thus, strong regional colocalisation was not always accompanied by substantial shared-variant posterior mass within the fine-mapped GWAS credible sets.

Integration of fine-mapped variants with molecular-QTL evidence The integration of credible-set variants, functional annotation, and molecular-QTL colocalisation highlighted several candidate mechanistic associations.

At the *GAK* region (GCST90473307, L004), the GWAS credible-set variant rs11248057 had a PIP of approximately 0.972.

An external dorsolateral prefrontal cortex sQTL associated with *GAK* showed strong colocalisation support ($PP_{H4} \approx 1.000$), with conditional credible-set posterior mass of approximately 0.998.

A histone-associated QTL at the same locus also showed strong colocalisation ($PP_{H4} \approx 0.989$) and substantial credible-set posterior support.

At the *SNCA* region, rs356219 was supported by GWAS fine-mapping in GCST90319903 locus L003 (PIP approximately 0.926).

An external posterior cingulate cortex splicing-QTL comparison involving *SNCA* showed $PP_{H4} \approx 0.997$, with high credible-set posterior support.

Similarly, rs356182 had a GWAS PIP of approximately 0.957 in GCST90476962 locus L001.

At this locus, a posterior cingulate cortex sQTL associated with *SNCA* showed $PP_{H4} \approx 0.991$ and credible-set posterior mass of approximately 0.931.

These results supported the prioritisation of candidate RNA-processing and regulatory associations in the chromosome 4 region.

Complex linkage disequilibrium and uncertainty at the MAPT region The chromosome 17q21.31 region containing *MAPT*, *KANSL1*, and related genes showed multiple strong molecular-QTL colocalisation comparisons.

However, this region contains extensive linkage disequilibrium and a complex inversion-associated haplotype structure.

Several external sQTL comparisons showed very high regional ABF colocalisation probabilities despite negligible conditional posterior mass within the GWAS credible sets.

For example, an external sQTL comparison for GCST90319903 locus L006 had $PP_{H4} \approx 0.999$, but its conditional credible-set posterior mass was approximately

$$M_{\mathrm{CS}|H4} \approx 1.5 \times 10^{-12}. \tag{19}$$

This discrepancy indicated that the variant-level interpretation of the regional ABF result was not consistent with the retained GWAS credible-set evidence.

Because `coloc.abf` assumes a single causal association signal per trait, such results require caution when several independent signals or complex LD patterns are present.

Signal-level molecular-QTL fine-mapping and multiple-signal colocalisation would provide a more appropriate follow-up for these regions.

3.8.5 Integrated biological interpretation

The combined analyses provided molecular context for GWAS-associated genomic regions by integrating fine-mapping, functional annotation, and colocalisation evidence.

Across Steps 09–11, 456 strong candidate trait/context comparisons were identified from 35,466 completed tests.

The results supported the prioritisation of candidate mechanisms involving RNA isoform usage, RNA splicing, gene expression, and chromatin-associated regulation, particularly within the *SNCA*, *GAK*, and 17q21.31 genomic regions.

Importantly, strong regional colocalisation was distinguished from credible-set-supported variant-level evidence.

The latter enabled more focused variant prioritisation, while fine-mapping uncertainty, LD complexity, context specificity, and single-signal model assumptions were retained in the interpretation.

Although these findings provide candidate molecular explanations for the GWAS signals, they do not independently establish causal variants, effector genes, or mediation of Parkinson's disease risk.

Table 22: Representative external molecular-QTL colocalisation findings integrated with GWAS credible-set variants. PIP denotes the GWAS posterior inclusion probability. CS mass denotes conditional H4 variant-posterior mass within the GWAS credible set. These results represent associations rather than experimentally validated causal mechanisms.

GWAS	Locus	QTL trait	Tissue	CS variant	PIP	PP_{H4}	CS mass
GCST90473307	L004	*GAK* sQTL	DLPFC	rs11248057	0.972	1.000	0.998
GCST90473307	L004	Histone-associated QTL	DLPFC	rs11248057	0.972	0.989	1.000
GCST90319903	L003	*SNCA* sQTL	PCC	rs356219	0.926	0.997	1.000
GCST90473307	L005	*SNCA* sQTL	PCC	rs356219	0.663	0.997	1.000
GCST90476961	L003	*SNCA* sQTL	PCC	rs356182	0.419	0.994	1.000
GCST90476962	L001	*SNCA* sQTL	PCC	rs356182	0.957	0.991	0.931

Further signal-level colocalisation, independent molecular datasets, and functional experimental validation will be necessary to establish the proposed biological mechanisms.

3.9 External benchmarking against Open Targets (workflow step 13)

3.9.1 Rationale and benchmark design

To evaluate the concordance of GWAS2Mech-derived genetic and molecular evidence with an established external resource, we compared the integrated results from Step 12 against Open Targets Platform release 26.09, released in September 2026 [3, 4, 11].

Open Targets integrates GWAS fine-mapping, molecular-QTL colocalisation, functional genomic annotations, and machine-learning-based Locus-to-Gene (L2G) prioritisation.

Unlike conventional gene-overlap comparisons, the present benchmark evaluated agreement at four levels:

1. GWAS study and genomic locus concordance;
2. Fine-mapped variant and credible-set overlap;
3. Candidate effector-gene prioritisation;
4. Molecular-QTL colocalisation evidence.

The comparison used the same GWAS Catalog study accessions wherever available.

Open Targets data were retrieved from the release-specific Parquet datasets containing GWAS credible sets, L2G predictions, colocalisation results, and study metadata.

The analysis used GRCh38 coordinates and allele-resolved variant identifiers.

GWAS2Mech genomic regions were compared with Open Targets credible sets using original GWAS lead variants, genomic overlap, and shared fine-mapped variants.

Because a GWAS2Mech region may contain multiple independent association signals, one-to-many region-to-credible-set matches were retained. A primary matching credible set was additionally selected for summary reporting.

The original clumping lead variant was distinguished from the highest-PIP variant and the lead variant of an individual fine-mapped credible set.

3.9.2 Fine-mapping concordance criteria

For comparable credible sets, variant membership was evaluated using the Jaccard similarity:

$$J_{\mathrm{CS}} = \frac{|\mathcal{C}_{\mathrm{GWAS2Mech}} \cap \mathcal{C}_{\mathrm{OT}}|}{|\mathcal{C}_{\mathrm{GWAS2Mech}} \cup \mathcal{C}_{\mathrm{OT}}|}. \tag{20}$$

Here, $\mathcal{C}_{\mathrm{GWAS2Mech}}$ and $\mathcal{C}_{\mathrm{OT}}$ represent the corresponding sets of fine-mapped variants.

Posterior inclusion probability concordance was evaluated using Spearman and Pearson correlations where a sufficient number of shared variants was available.

Comparisons were interpreted according to the fine-mapping method used by Open Targets, including SuSiE-inf and PICS.

PICS-derived credible sets were not assumed to be directly equivalent to SuSiE-RSS association components.

3.9.3 Effector-gene and molecular-QTL comparison criteria

Gene prioritisation was compared using Ensembl gene identifiers.

For each matched Open Targets credible set, we extracted the L2G prediction score, gene rank, and available colocalisation evidence.

The L2G score was interpreted as a model-derived prioritisation score rather than an experimentally measured probability of causality.

Molecular-QTL concordance required a corresponding gene identifier and an Open Targets colocalisation result meeting the predefined posterior threshold:

$$PP_{H4}^{\mathrm{OT}} \geq 0.80. \tag{21}$$

Open Targets derives molecular colocalisation from fine-mapped credible-set overlaps using COLOC-PIP and eCAVIAR-related statistics.

These measures were not assumed to be numerically interchangeable with the SuSiE-based or approximate-Bayes-factor colocalisation analyses in GWAS2Mech.

Consequently, the benchmark evaluated the consistency of molecular-gene evidence, not direct equality of posterior probabilities.

A same-gene match was distinguished from a fully matched molecular mechanism because the corresponding QTL types, tissues, cell populations, and conditions were not necessarily identical.

Absence from Open Targets was not interpreted as evidence against a biological association or as proof of novelty.

3.9.4 Study matching and data integration

To evaluate the concordance between GWAS2Mech and an established genetic evidence integration framework, we compared our results with Open Targets Platform release 26.09 [3, 4].

The comparison integrated the outputs of GWAS2Mech Steps 05–12 with four Open Targets datasets: credible sets, Locus-to-Gene (L2G) predictions, molecular-QTL colocalisation, and study metadata.

Comparisons were restricted to the same GWAS Catalog study accession where supported by the external resource. Variant identifiers were harmonised using GRCh38 chromosome, position, reference allele, and alternative allele.

GWAS2Mech genomic regions and Open Targets credible sets were compared using lead-variant concordance, genomic overlap, credible-set variant membership, and posterior inclusion probabilities (PIPs).

Because a GWAS2Mech genomic region could contain multiple association components, all supported region-to-credible-set matches were retained.

Gene-level comparisons used Ensembl identifiers, Open Targets L2G scores and ranks, and molecular-QTL colocalisation evidence.

The analyses distinguished two different outputs:

$$PP_{H4}^{\mathrm{GWAS2Mech}} = P(H_4 \mid D_{\mathrm{GWAS}}, D_{\mathrm{QTL}}) \tag{22}$$

and

$$S_{\mathrm{L2G}}^{\mathrm{OT}} \in [0, 1], \tag{23}$$

where $PP_{H4}^{\mathrm{GWAS2Mech}}$ represents Bayesian support for a shared association signal, whereas $S_{\mathrm{L2G}}^{\mathrm{OT}}$ denotes a machine-learning-based effector-gene prioritisation score.

These measurements answer different questions and were therefore reported side by side without treating their numerical magnitudes as directly interchangeable.

For molecular-QTL comparisons, the maximum retrieved Open Targets same-gene colocalisation posterior was recorded separately.

An Open Targets molecular result was considered strong when its reported H4 support was at least 0.80.

Gene concordance did not automatically imply agreement on the molecular-QTL modality, tissue, cell type, or experimental condition.

3.9.5 Overall benchmarking results

GWAS2Mech identified 32 study-specific genomic regions containing 69 SuSiE-RSS credible sets.

Open Targets provided 19 credible sets associated with the corresponding GWAS Catalog accessions.

Fourteen GWAS2Mech regions matched at least one Open Targets credible set.

Accounting for multiple credible sets within individual regions resulted in 17 supported region–credible-set comparison pairs.

Of the 14 primary regional matches, 13 were supported by agreement with an original Step 05 GWAS lead variant.

Twelve GWAS2Mech regions had no same-study Open Targets credible-set coverage, while six additional regions remained unmatched despite study-level coverage.

Two Open Targets credible sets had no corresponding GWAS2Mech regional match.

The observed regional concordance was therefore 14/20 (70.0%) among GWAS2Mech regions in studies represented by Open Targets.

This statistic describes agreement with the selected database, not absolute biological precision or causal-discovery accuracy.

Table 23: Summary of the Open Targets comparison. All counts refer to the completed Step 13 run using release 26.09.

Measurement	Result
GWAS2Mech genomic regions	32
GWAS2Mech SuSiE credible sets	69
Open Targets credible sets	19
Matched GWAS2Mech regions	14
Supported region–OT credible-set pairs	17
Original Step 05 lead agreement	13/14
Pairs with shared credible-set variants	10/17
SuSiE component-pair comparisons	55
Component pairs sharing CS variants	16
Regions without same-study OT coverage	12
Unmatched regions with OT study coverage	6
Unmatched Open Targets credible sets	2

3.9.6 Detailed locus and variant concordance

For each supported region–credible-set pair, we compared the highest-PIP variant from GWAS2Mech with the highest-ranked variant from Open Targets.

Credible-set overlap was quantified using the Jaccard index:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}. \tag{24}$$

Posterior inclusion probability concordance was evaluated using Spearman correlation where sufficient shared variants were available.

Seven of the 17 supported pairs shared the same highest-ranked variant.

Ten pairs shared at least one variant belonging to the respective 95% credible-set collections.

At the individual component level, 16 of 55 GWAS2Mech–Open Targets credible-set combinations contained at least one shared variant.

The median PIP Spearman correlation was 0.687 among 12 comparable pairs.

Table 24 provides the complete 17-pair comparison.

Table 24: All 17 supported GWAS2Mech region–Open Targets credible-set pairs. Variant identifiers use GRCh38 chromosome:position:reference:alternate. GWAS2M top and OT top denote the highest-ranked fine-mapped variants, not necessarily the original clumping lead. Shared CS represents the number of overlapping variants in the aggregated 95% credible-set comparison. J is the credible-set Jaccard index and ρ is PIP Spearman correlation. A regional match does not necessarily imply a shared causal signal.

GWAS accession	Locus	Open Targets credible-set ID	GWAS2Mech top variant	Open Targets top variant	Shared CS	J_{CS}	ρ_{PIP}
GCST90473307	L008	2bca8efd63c5c43f51763138f70911a8	17:45582160:C:T	17:45825175:A:G	0	0.000	-0.107
GCST90042681	L002	df38b480d3423d85a3174b5ab87bf636	17:46026338:G:A	17:45771384:T:C	0	0.000	0.726
GCST90042681	L002	5d477c82de053b6cd17ae7319213f6b5	17:46026338:G:A	17:46150803:C:T	0	0.000	0.874
GCST90473307	L004	755b0e09f64f692ff482f007d74f063c	4:912343:C:G	4:912343:C:G	1	0.500	–
GCST90473307	L005	5afaebd1003d5a6887b53986404706b1	4:89716450:G:A	4:89716450:G:A	3	0.037	0.647
GCST90473307	L001	79ee1ed847bc5971af8016a077d5a1a3	1:155236376:C:T	1:155236376:C:T	4	0.444	0.983
GCST90475830	L002	f769e78d138b57a328a5360908dd7c3c	4:89704947:G:C	4:89704947:G:C	12	0.750	0.967
GCST90476961	L003	266108143548f1b5a7f287394e592129	4:89704960:G:A	4:89704960:G:A	7	0.068	0.393
GCST90476962	L001	7a6a0bc08aff5ecd1ef1ae5f4e689b0d	4:89704960:G:A	4:89704960:G:A	1	0.009	0.800
GCST90475830	L004	4ffc339e13fdba18e19cd608c1adc2f2	17:45677566:ATT:A	17:45673544:C:T	4	0.001	0.018
GCST90475830	L004	047eb24e21cba273144c971b45df580a	17:45677566:ATT:A	17:46115202:C:T	1	0.001	-0.011
GCST90473307	L002	0b95879e8f1321d06f7efce72d1cb5d5	2:168257501:T:C	2:168309772:T:C	6	0.857	0.821
GCST90475830	L001	628e27772029d446344bdc91d43d2bd5	2:168286525:G:A	2:168286525:G:A	5	0.312	0.622
GCST90476961	L001	8e46c2546ea4867dc829e47c8da240ad	2:62290665:A:G	2:61458142:T:C	0	0.000	–
GCST90476961	L001	6096e29792a49dbd45f0c68ad644740c	2:62290665:A:G	2:61908206:A:G	0	0.000	–
GCST90473307	L006	d80e59124cc5ac317696a94a6bbb0dd4	12:39429746:T:G	12:40340400:G:A	0	0.000	–
GCST90475830	L003	726c98679586f0c7d9ebe0b29a9bed2d	12:39930140:A:C	12:40340400:G:A	0	0.000	–

3.9.7 Gene-level molecular evidence concordance

The integrated Step 12 analysis contained 1,601 locus–gene records.

Sixty-nine of these records had strong GWAS2Mech molecular-QTL colocalisation support ($PP_{H4} \geq 0.80$).

Forty-two strong locus–gene records occurred at regions matched to Open Targets credible sets.

Among these 42 records, eleven had corresponding Open Targets L2G predictions and seven were ranked first.

Eleven strong records also showed same-gene molecular-QTL support in Open Targets above the 0.80 threshold.

The full integrated candidate table (Table 25) presents all 42 strong matched locus–gene records, their representative GWAS2Mech molecular-association variants, QTL modalities, biological contexts, and both resources' evidence scores.

Table 25: Complete set of 42 strong GWAS2Mech locus–gene records at regions matched to Open Targets credible sets. GWAS2M H4 denotes the highest GWAS2Mech colocalisation posterior for the listed gene and study-specific locus. The variant identifies the reported GWAS variant associated with that molecular comparison; it is not necessarily the locus lead or a member of the 95% credible set. OT L2G gives the Locus-to-Gene score followed by its rank in parentheses. OT H4 is the maximum retrieved same-gene molecular colocalisation posterior at the matched OT credible set(s), potentially from a different molecular context. Dashes indicate unavailable evidence. Scores are rounded; 1.0000 does not indicate mathematical certainty. DLPFC, dorsolateral prefrontal cortex; PCC, posterior cingulate cortex; TCM, central memory T cell; TEM, effector memory T cell.

GWAS accession	Locus	Gene	GWAS2Mech variant	QTL	Context	GWAS2M H4	OT L2G (rank)	OT H4
GCST90473307	L004	*IDUA*	4:912343:C:G	exonQTL	breast	0.9999	–	0.9997
GCST90473307	L004	*GAK*	4:912343:C:G	sQTL	DLPFC	0.9996	0.856 (R1)	0.9997
GCST90473307	L005	*SNCA*	4:89716450:G:A	isoQTL	adipose	0.9978	0.929 (R1)	0.9978
GCST90473307	L001	*SLC50A1*	1:155236376:C:T	isoQTL	visceral adipose	0.9947	0.386 (R2)	0.9971
GCST90475830	L002	*SNCA*	4:89725735:G:A	sQTL	anterior caudate	0.9944	0.872 (R1)	0.9848
GCST90476961	L003	*SNCA*	4:89704960:G:A	sQTL	PCC	0.9942	0.921 (R1)	0.9956
GCST90476962	L001	*SNCA*	4:89704960:G:A	sQTL	PCC	0.9913	0.368 (R1)	0.9574
GCST90476962	L001	*ENSG00000251095*	4:89704960:G:A	exonQTL	suprapubic skin	0.9824	–	0.9725
GCST90042681	L002	*ARL17A*	17:46111701:A:G	eQTL	sigmoid colon	0.9763	0.071 (R3)	0.8262
GCST90476961	L003	*ENSG00000251095*	4:89704960:G:A	exonQTL	suprapubic skin	0.9655	–	0.9469
GCST90473307	L004	*TMEM175*	4:945299:C:T	pQTL	DLPFC	0.8870	0.088 (R2)	0.9809
Other strong records with L2G support, but without strong same-gene OT molecular H4								
GCST90475830	L004	*KANSL1*	17:45922040:A:G	isoQTL	cerebellum	0.9657	0.203 (R1)	0.6609
GCST90042681	L002	*KANSL1*	17:46111701:A:G	exonQTL	transverse colon	0.9296	0.172 (R1)	0.4741
GCST90475830	L004	*LRRC37A*	17:45726951:A:G	isoQTL	oesophagus (muscularis)	0.8417	0.051 (R3)	0.6162
Other strong GWAS2Mech candidates at matched OT regions								
GCST90473307	L008	*KANSL1*	17:45969671:T:A	sQTL	PCC	1.0000	–	0.7468
GCST90042681	L002	*LRRC37A4P*	17:46063913:A:T	eQTL	floor-plate progenitor	0.9999	–	0.4344
GCST90473307	L004	*DGKQ*	4:912343:C:G	exonQTL	skin	0.9998	–	–
GCST90473307	L008	*PLEKHM1*	17:46006582:T:C	sQTL	anterior caudate	0.9989	–	0.2738
GCST90042681	L002	*MAPT*	17:46111701:A:G	exonQTL	muscle	0.9971	–	0.5843
GCST90042681	L002	*ENSG00000262879*	17:46111701:A:G	exonQTL	prostate	0.9968	–	0.6143
GCST90042681	L002	*ENSG00000267198*	17:46111701:A:G	isoQTL	uterus	0.9944	–	0.4172
GCST90042681	L002	*LRRC37A2*	17:47236556:G:A	eQTL	substantia nigra	0.9916	–	0.5630
GCST90473307	L008	*MAPT*	17:45982034:G:A	sQTL	DLPFC	0.9867	–	0.3333
GCST90473307	L008	*ARL17B*	17:46122825:C:T	sQTL	anterior caudate	0.9825	–	0.2819
GCST90473307	L008	*LRRC37A2*	17:45974038:G:A	sQTL	PCC	0.9765	–	0.3781
GCST90042681	L002	*ARL17B*	17:46054002:T:C	eQTL	$CD4^+$ TCM	0.9740	–	0.4655
GCST90042681	L002	*LRRC37A*	17:46111701:A:G	exonQTL	DLPFC	0.9681	–	0.7545
GCST90473307	L005	*SNCA-AS1*	4:89828874:C:T	eQTL	pituitary	0.9564	–	–
GCST90476961	L003	*SNCA-AS1*	4:89828874:C:T	eQTL	pituitary	0.9531	–	–
GCST90475830	L004	*ARL17B*	17:45726951:A:G	eQTL	transverse colon	0.9246	–	0.6603

Continued on next page

Table continued from previous page

GWAS accession	Locus	Gene	GWAS2Mech variant	QTL	Context	GWAS2M H4	OT L2G (rank)	OT H4
GCST90473307	L008	*KANSL1-AS1*	17:45616172:G:T	exonQTL	spleen	0.9203	–	0.4419
GCST90473307	L008	*LRRC37A4P*	17:45605027:A:G	sQTL	anterior caudate	0.9126	–	0.3245
GCST90476962	L001	*SNCA-AS1*	4:89827044:T:A	eQTL	pituitary	0.9091	–	–
GCST90473307	L008	*LINC02210-CRHR1*	17:45609902:G:A	sQTL	anterior caudate	0.8996	–	0.2610
GCST90473307	L008	*FAM215B*	17:46282948:A:G	eQTL	floor-plate progenitor	0.8905	–	0.2742
GCST90473307	L008	*LINC02210*	17:45765107:G:A	sQTL	PCC	0.8729	–	0.5998
GCST90475830	L004	*LRRC37A2*	17:45726951:A:G	exonQTL	anterior cingulate	0.8639	–	0.4985
GCST90042681	L002	*NSFP1*	17:45994485:G:A	sQTL	PCC	0.8590	–	0.2686
GCST90475830	L004	*LINC02210*	17:45786127:C:CT	eQTL	$CD8^+$ TEM	0.8463	–	0.4644
GCST90473307	L008	*ARL17A*	17:45954230:A:G	isoQTL	putamen	0.8387	–	0.2922
GCST90475830	L004	*KANSL1-AS1*	17:45786127:C:CT	eQTL	$CD4^+$ T cell	0.8337	–	0.5076
GCST90473307	L008	*LRRC37A*	17:45616172:G:T	isoQTL	skin	0.8168	–	0.3560

3.9.8 Biological interpretation of concordant genes

The comparison highlighted several candidate effectors with evidence from both pipelines.

At GCST90473307 locus L004, *GAK* showed a GWAS2Mech colocalisation posterior of 0.9996, associated with a dorsolateral prefrontal cortex splicing QTL.

Open Targets ranked *GAK* first with an L2G score of 0.856 and reported same-gene molecular-QTL colocalisation support of 0.9997.

The allele-resolved GWAS2Mech variant reported for this molecular comparison was `4:912343:C:G` (GRCh38).

At GCST90473307 locus L005, *SNCA* showed a GWAS2Mech isoform-QTL colocalisation posterior of 0.9978, associated with adipose tissue.

The corresponding Open Targets L2G score was 0.929, with rank one, and the maximum same-gene molecular colocalisation posterior was 0.9978.

Additional *SNCA* associations were supported in GCST90475830, GCST90476961, and GCST90476962.

At GCST90473307 locus L001, *SLC50A1* showed a GWAS2Mech isoform-QTL posterior of 0.9947, while Open Targets ranked the gene second with an L2G score of 0.386.

The external same-gene molecular colocalisation posterior was 0.9971.

The *IDUA* and *TMEM175* associations at locus L004 also demonstrated molecular-QTL evidence, although their L2G prioritisation differed from that of *GAK*.

These observations demonstrate that multiple genes within the same genomic region may have supporting molecular associations, while the gene prioritisation model ranks them differently.

Importantly, the reported molecular contexts were derived from the GWAS2Mech analysis. The corresponding Open Targets molecular evidence was matched at the gene level and did not necessarily involve the same tissue or molecular phenotype.

3.9.9 Differences in effector-gene prioritisation

The benchmark also revealed differences between the candidate genes prioritised by GWAS2Mech molecular-QTL evidence and Open Targets L2G predictions.

For example, *KANSL1* at GCST90475830 locus L004 showed strong GWAS2Mech isoform-QTL colocalisation ($PP_{H4} = 0.9657$).

Open Targets ranked *KANSL1* first with an L2G score of 0.203, but its maximum retrieved same-gene molecular-QTL H4 was 0.661, below the predefined strong colocalisation threshold.

Similarly, the complex chromosome 17 region containing *MAPT*, *LRRC37A*, *ARL17A*, *ARL17B*, and related genes showed several strong GWAS2Mech molecular associations without equally strong matching Open Targets molecular evidence.

These differences were not interpreted as failures of either method or as proof of novel biological mechanisms.

Potential explanations include different LD references, fine-mapping models, molecular-QTL datasets, gene-biotype eligibility, and tissue-specific coverage.

Particular caution was applied to the 17q21.31 region because of its complex linkage disequilibrium and multiple association signals.

Where the GWAS2Mech result originated from the single-signal `coloc.abf` analysis, the accompanying credible-set posterior support and assumption warnings were retained for follow-up interpretation.

3.9.10 Candidate categories and follow-up evidence

The 1,601 integrated locus–gene records were categorised according to the available external evidence.

Seventeen records, representing 11 distinct genes, exhibited same-gene molecular-QTL concordance with Open Targets.

Fifteen records had Open Targets L2G support but did not satisfy the matching molecular-QTL criterion.

A further 643 records represented alternative candidates at Open Targets-matched regions, while 918 records were classified as unresolved owing to limited coverage or mapping.

Eight records had strong GWAS2Mech molecular support at regions without corresponding same-study Open Targets locus matches.

These numbers describe evidence categories rather than numbers of newly discovered genes or loci.

Table 26: Open Targets evidence-classification results. Rows represent study-specific locus–gene records, not unique genes.

Category	Records
Concordant molecular-QTL evidence	17
L2G-supported, differing molecular support	15
Alternative candidate at matched region	643
Unmatched region, external validation required	8
Coverage or mapping unresolved	918
Total	1,601

3.9.11 Open Targets signals not recovered by GWAS2Mech

Two Open Targets credible sets from GCST90473307 were unmatched to GWAS2Mech genomic regions.

These involved lead variants at GRCh38 chromosome 3 position 128,656,141 and chromosome 12 position 37,686,151.

The corresponding GWAS association regions had previously been flagged during reference-panel harmonisation because significant variants could not be matched to a compatible reference-panel representation.

Table 27: Open Targets credible-set lead variants not recovered as GWAS2Mech genomic loci.

GWAS accession	OT lead variant	Chromosome
GCST90473307	`3:128656141:C:T`	3
GCST90473307	`12:37686151:T:A`	12

These unmatched associations should not be interpreted as evidence against the original GWAS signals.

Instead, they highlight reference-harmonisation and fine-mapping coverage limitations that could be addressed through alternative variant mapping or reference-panel strategies.

3.9.12 Benchmarking conclusions and limitations

The external comparison demonstrated concordance between GWAS2Mech and Open Targets for selected genomic regions, fine-mapped variants, candidate effector genes, and molecular-QTL associations.

The strongest gene-level agreement was observed at the *GAK* and *SNCA* regions, with additional support for *SLC50A1*, *IDUA*, and *TMEM175*.

However, genomic-region matches were distinguished from credible-set and causal-signal agreement.

Furthermore, differences between the molecular-QTL colocalisation methods, biological contexts, and gene-prioritisation models limited direct score comparison.

Open Targets and GWAS2Mech may also incorporate overlapping GWAS or molecular-QTL datasets, so agreement represents computational concordance rather than independent experimental replication.

The local Open Targets Parquet datasets were readable, but full release completeness was not independently verified against the remote release manifest.

The analysis demonstrates that GWAS2Mech can reproduce selected established genetic and molecular associations while identifying additional candidate genes and biological contexts for subsequent validation.

The results do not establish previously unknown causal genes, disease mechanisms, or functional effects without further evidence.

3.10 Candidate-gene prioritisation and pathway analysis (workflow step 14)

3.10.1 Rationale and study aggregation

We evaluated candidate gene functions after integrating fine-mapped GWAS associations, variant annotations, and molecular-QTL colocalisation evidence in Step 12. The integrated table contained 1,601 study-specific locus–gene records, of which 1,137 records had resolvable gene identifiers and represented 395 unique genes. The analyses encompassed different GWAS accessions, locus assignments, molecular-QTL types, tissues, and cell types. Candidate evidence was pooled across the represented GWAS studies to construct a single disease-level gene ranking; individual studies did not each contribute an independent vote to the pathway test. For enrichment, a gene was counted at most once per gene set irrespective of the number of study-specific associations or molecular contexts. Independent per-study or per-cohort pathway enrichment was not performed in this run.

3.10.2 Candidate-gene prioritisation and quality control

For each unique Ensembl gene identifier, Step 14 integrated supported genetic–molecular associations from bulk molecular-QTL analysis (Step 09), cell-type QTL analysis (Step 10), and additional molecular-QTL datasets (Step 11). The primary gene set required a colocalisation posterior $PP_{H4} \geq 0.80$ in at least one association passing the source-specific support checks. Step 09/10 associations required a documented linkage to a fine-mapped GWAS signal or credible set; Step 11 approximate-Bayes-factor associations required candidate-specific credible-set posterior support of at least 0.50 without a recorded ABF assumption warning. The run did not impose the optional strict region-wide fine-mapping QC requirement, and corresponding fine-mapping warnings therefore remain important limitations.

The final score was a predefined heuristic calculated only from QC-supported molecular associations:

$$S_g = H_g^{\mathrm{QC}} + 0.15R_g + 0.08E_g + 0.08\min(B_g, 3), \tag{25}$$

where H_g^{QC} is the maximum colocalisation PP_{H4} among supported associations; R_g equals one if at least one such association was robust to the colocalisation-prior sensitivity settings and zero otherwise; E_g is the number of distinct qualifying evidence-source categories (bulk, cell-type, and additional molecular-QTL evidence); and B_g counts distinct genomic-region blocks carrying supported associations. The source-specific counts of QTL measurements or overlapping GWAS studies were not added as repeated votes. Genes were ordered by QC-supported status, descending supported-region-block count, the heuristic score, and subsequent tie-breakers. The score is not a calibrated posterior probability that a gene is causally responsible for Parkinson's disease. Open Targets evidence was appended after ranking and was not an input to Eq. 25.

Twenty-four unique genes qualified for the primary QC-supported strong set, compared with 27 under the unfiltered $PP_{H4} \geq 0.80$ definition and 31 under a suggestive $PP_{H4} \geq 0.50$ threshold. Nineteen primary genes satisfied the prior-robust criterion. The entire ranked 395-gene catalogue, including genes outside these prespecified groups, should be provided as a supplementary table generated from `Step14_Gene_Ranking.tsv`.

3.10.3 Gene biotypes and interpretation of noncoding candidates

A noncoding GWAS variant is not equivalent to a noncoding gene: a variant outside a protein-coding exon may regulate expression or splicing of a protein-coding gene. Separately, noncoding RNA genes can exert indirect pathway effects through transcriptional or post-transcriptional regulation of other genes. The present analysis retained Ensembl gene identifiers without systematically assigning verified GENCODE biotypes or reconstructing directed noncoding-RNA regulatory edges. Therefore, pathway non-membership for an RNA gene cannot be interpreted as biological inactivity. Biotype-specific numbers and assignments should only be added after checking the Ensembl/GENCODE annotation release.

3.10.4 Pathway over-representation analysis

The 24 deduplicated primary genes were submitted to g:Profiler for Gene Ontology Biological Process, Reactome, KEGG, and WikiPathways analyses. The online analysis used the g:Profiler

Table 28: All 24 QC-supported candidate genes in Step 14, ordered by the final heuristic rank. $H_4^{\max}$ is the maximum displayed colocalisation posterior across the gene's associations; for genes whose overall best result fails QC it may differ from H_g^{QC} used in the score. These are candidate genes, not independently established causal effectors.

Rank	Gene (Ensembl ID)	$H_4^{\max}$	Score S_g
1	SNCA (ENSG00000145335)	0.997807	1.46781
2	GAK (ENSG00000178950)	0.999602	1.38960
3	ARL17B (ENSG00000228696)	0.996536	1.38654
4	LRRC37A2 (ENSG00000238083)	0.996016	1.38602
5	LRRC37A (ENSG00000176681)	0.996016	1.38602
6	ENSG00000267198	0.994419	1.38442
7	KANSL1 (ENSG00000120071)	0.999985	1.35674
8	KANSL1-AS1 (ENSG00000214401)	0.962364	1.35236
9	LRRC37A4P (ENSG00000214425)	0.999939	1.30994
10	IDUA (ENSG00000127415)	0.999909	1.30991
11	DGKQ (ENSG00000145214)	0.999771	1.30977
12	MAPT (ENSG00000186868)	0.997851	1.30785
13	ENSG00000262879	0.996791	1.30679
14	ARL17A (ENSG00000185829)	0.996016	1.30602
15	SLC50A1 (ENSG00000169241)	0.995111	1.30511
16	ENSG00000251095	0.982408	1.29241
17	NSFP1 (ENSG00000260075)	0.972259	1.28226
18	SNCA-AS1 (ENSG00000247775)	0.967495	1.27749
19	STK39 (ENSG00000198648)	0.957714	1.26771
20	GCH1 (ENSG00000131979)	0.939342	1.09934
21	NSF (ENSG00000073969)	0.934678	1.09468
22	FAM215B (ENSG00000232300)	0.934409	1.09441
23	LINC02210 (ENSG00000204650)	0.872938	1.00634
24	FMNL1-AS1 (ENSG00000233175)	0.820767	0.98077

annotated-human-gene background, restricted reported terms to size 5–500 with at least two overlapping genes, and applied a false-discovery rate cutoff of 0.05. This reference background was exploratory and did not establish the complete molecular-QTL assay-eligible gene universe; thus, the enrichment findings require sensitivity analysis with an assay-appropriate background before being designated publication-final.

Ninety pathway terms satisfied the run's criteria in the primary 24-gene set, many of which represent overlapping or hierarchically related biological processes. Representative terms included synaptic vesicle trafficking, dopamine biosynthesis, responses to cellular oxidative stress, and myeloid/glial activation (Table 29). The pathway tests measure over-representation of annotated genes and do not establish whether a pathway is activated, inhibited, or causally changed by a risk allele.

Table 29: Representative enriched terms for the pooled 24-gene primary set in the exploratory annotated-genome analysis. q denotes the adjusted P-value returned by g:Profiler. Contributing genes are annotated pathway members, not inferred direct causal mediators.

Pathway or process	q	Contributing input genes
Secretase involvement in neurodegenerative diseases	0.002152	SNCA, MAPT
Synaptic vesicle recycling	0.003226	DGKQ, SNCA, GAK
Synaptic vesicle endocytosis	0.003226	DGKQ, SNCA, GAK
Dopamine biosynthetic process	0.003226	GCH1, SNCA
Cellular response to chemical stress	0.003226	GCH1, SNCA, MAPT, STK39
Macrophage activation	0.003226	SNCA, MAPT, STK39
Synaptic vesicle cycle	0.009621	DGKQ, SNCA, GAK
Cellular response to oxidative stress	0.009786	GCH1, SNCA, MAPT
Microglial cell activation	0.009786	SNCA, MAPT

The pathway analyses were performed on pooled disease-level gene sets rather than independently within each GWAS cohort. Sensitivity enrichments included the 27-gene unfiltered strong set (84 enriched terms), the 19-gene prior-robust set (66 terms), and the 31-gene suggestive set (78 terms). The 24-gene QC-supported primary set and these sensitivity sets overlap and should not be summed as independent discoveries.

3.10.5 Comparison of derived pathways with Open Targets

Open Targets benchmarking in Step 13 linked GWAS2Mech genomic regions to corresponding external study-specific credible sets. For the pathway-level comparator, Step 14 extracted each valid `OT_L2G_TOP1_GENE` identifier from the Step 13 gene comparison and deduplicated the identifiers across the represented credible sets. This generated six distinct comparison genes: *KANSL1* (ENSG00000120071), *SNCA* (ENSG00000145335), *GBA1* (ENSG00000177628), *GAK* (ENSG00000178950), *STK39* (ENSG00000198648), and ENSG00000256762. These are six distinct top-ranked L2G genes across the selected external credible sets, not the complete Open Targets Parkinson's disease effector catalogue.

Table 30: Six distinct Open Targets top-ranked L2G genes extracted from Step 13. The external score is credible-set-specific and is therefore not presented as one pooled value per gene.

Gene	Ensembl ID
KANSL1	ENSG00000120071
SNCA	ENSG00000145335
GBA1	ENSG00000177628
GAK	ENSG00000178950
STK39	ENSG00000198648
Unresolved symbol	ENSG00000256762

Gene-specific Open Targets L2G scores are credible-set dependent, and are documented separately in Step 13.

We submitted these six identifiers to the same enrichment sources, yielding 48 terms meeting the chosen criteria. Comparing term identifiers with the 90 terms identified from the 24 GWAS2Mech genes yielded 23 terms in common, 67 only in the GWAS2Mech enrichment, and 25 only in the six-gene Open Targets enrichment. The pathway-set Jaccard similarity was

$$J_{\text{pathways}} = \frac{|P_{\text{GWAS2Mech}} \cap P_{\text{OT}}|}{|P_{\text{GWAS2Mech}} \cup P_{\text{OT}}|} = \frac{23}{23 + 67 + 25} = 0.20. \tag{26}$$

Because the two input gene sets differ greatly in size (24 versus six), this comparison represents descriptive annotation overlap, not a controlled comparison of causal-gene inference accuracy or pathway discovery performance. Absence of an L2G prediction may additionally reflect protein-coding gene eligibility and model filtering.

3.10.6 Leave-one-gene-out and leave-one-region-out sensitivity

We quantified pathway dependence by repeating enrichment after each of 24 individual gene removals and each of six genomic-region-block removals. During region removal, a gene was

retained if its qualifying evidence was also available in another genomic block. Thirty of 30 perturbation runs completed successfully. We followed both significant and nonsignificant baseline pathway terms, including pathways absent from a perturbed result, rather than imputing unreported P-values.

Table 31 summarises representative baseline-significant terms. The 90 original primary terms included no term that remained significant after every gene removal and every region removal. Removing the chromosome 17 block labelled `chr17_block002` reduced the primary list from 24 to nine genes, demonstrating a marked dependence on the contribution of that complex association region. The sensitivity fractions reflect the proportion of computational deletion tests preserving enrichment significance; they do not represent replication probabilities.

Table 31: Selected pathway sensitivity results. Numerators indicate perturbations retaining FDR significance.

Pathway	Region LOO	Gene LOO
Secretase involvement in neurodegenerative diseases	4/6	22/24
Synaptic vesicle recycling	4/6	23/24
Synaptic vesicle endocytosis	4/6	23/24
Dopamine biosynthetic process	4/6	22/24
Cellular response to chemical stress	5/6	23/24
Macrophage activation	5/6	23/24
Cellular response to oxidative stress	5/6	23/24
Microglial cell activation	4/6	22/24

3.10.7 Interpretation and remaining mechanistic questions

The integrated analysis provided a reproducible, evidence-weighted ranking of 395 candidate genes, including 24 with at least one QC-supported strong molecular association, and identified over-represented biological annotations in the pooled gene set. The findings highlight neuronal vesicle dynamics, catecholamine metabolism, cellular-stress responses, and immune/glial processes, but do not establish whether changing a candidate gene alters those pathways or changes Parkinson's disease risk. Determining the direction of action will require credible-set-resolved regulatory links, molecular effect direction, independently supported gene–pathway relationships, and, where feasible, gene perturbation experiments or suitably relevant perturbational datasets. Protein association networks may provide complementary context but cannot independently validate causality or represent noncoding-RNA regulatory effects comprehensively.

3.11 Protein-association networks and functional context (workflow step 15)

3.11.1 Rationale and analytical strategy

Following the integrated candidate-gene prioritisation and exploratory pathway analysis in Step 14 (Section 3.10), we evaluated whether proteins corresponding to prioritised Parkinson's disease candidate genes exhibited nonrandom network connectivity and whether curated or predicted protein associations could place the pathway findings into a coherent molecular context. Importantly, Step 15 was designed as a *protein-association and network-context analysis*, not as a causal gene validation, gene perturbation, or drug-discovery analysis. We analysed the

same pooled disease-level priority gene set used in Step 14; we did not construct independent physical-interaction networks for each GWAS cohort.

The primary input comprised 24 distinct QC-supported genes in the `CORE_STRONG` set, with additional prespecified sensitivity and comparison sets from Step 14. The external comparator was a set of six distinct Open Targets top-ranked locus-to-gene (L2G) genes recovered in Steps 13–14: *KANSL1*, *SNCA*, *GBA1*, *GAK*, *STK39*, and ENSG00000256762. The last identifier was resolved by the STRING mapping service to protein *STH*; *GBA1* was displayed under STRING's preferred protein label *GBA*. Open Targets-only nodes were retained as external comparators and were excluded from the *primary GWAS2Mech-only* network.

Human protein identifiers were resolved with the STRING v12.5 API using NCBI taxonomy identifier 9606 [7]. We submitted stable Ensembl gene identifiers, with a gene-symbol fallback when required, and audited gene biotypes against Ensembl annotations. Networks were requested at a minimum STRING confidence score of 700/1000, separately for `network_type=physical` and `network_type=functional`. The unexpanded networks included only seed proteins; an exploratory network additionally allowed up to 20 first-order STRING connector proteins (`add_nodes=20`). Connector nodes were explicitly distinguished from genes supported by GWAS2Mech genetic evidence. We computed degree, betweenness centrality, PageRank, connected components and network communities using NetworkX. These metrics summarise network topology and should not be interpreted as molecular effect sizes or probabilities of causality.

The STRING `ppi_enrichment` endpoint was also queried independently for six prespecified gene sets, using the organism-level STRING reference network as the background. The test evaluates whether the submitted proteins are more connected than expected under the STRING background interaction distribution; it is *not* the same calculation as counting edges in the `physical`-only network. We additionally applied Benjamini–Hochberg (BH) correction to the six returned P-values. The six tested sets overlap and should not be treated as independent biological replications. BioGRID experimental interaction checking was supported by the code but *was not performed* because this execution did not supply BioGRID credentials; no BioGRID-supported interactions are claimed in the following results [8].

3.11.2 Gene-biotype audit and protein mapping

The primary 24-gene set contained 14 Ensembl-annotated protein-coding genes and 10 non-protein-coding genes, with no unknown gene biotypes in this set. All 14 protein-coding primary genes were retained as mapped STRING proteins in the audited primary network. A preliminary mapping stage initially returned 15/24 protein assignments, but one gene-symbol fallback assigned the noncoding locus *LINC02210* to the protein label *CRHR1-2*; the subsequent gene-biotype audit prevented that potentially misleading assignment from entering the primary protein-only network. Thus, the biologically interpretable primary physical network contained *14 mapped proteins*, not 15. The mapped primary genes were *NSF*, *KANSL1*, *IDUA*, *GCH1*, *DGKQ*, *SNCA*, *SLC50A1*, *LRRC37A*, *GAK*, *ARL17A*, *MAPT*, *STK39*, *ARL17B*, and *LRRC37A2*. The ten excluded noncoding candidates were preserved in the mapping audit and in the gene–pathway evidence handoff, rather than being classified as biologically inactive.

The combined direct-network analysis contained the 14 primary mapped proteins plus the

Table 32: Prespecified parameters and resources for Step 15. The score threshold applied to STRING network retrieval and PPI-enrichment queries; physical association scores are not direct experimental binding probabilities.

Parameter	Implemented value
Primary genes	24 QC-supported Step 14 genes (`CORE_STRONG`)
External comparison	Six deduplicated Open Targets L2G top-ranked genes
Organism	*Homo sapiens*, NCBI taxonomy 9606
Protein association source	STRING v12.5, version-specific API
Minimum association score	700/1000 (high confidence)
Network representations	Direct physical; direct functional; expanded physical; expanded functional
Exploratory connector limit	20 first-order proteins
PPI-enrichment reference	STRING organism-level protein-association background
Multiple-testing adjustment	BH correction over the six tested gene-set PPI P-values
Annotation and pathway mapping	Ensembl gene biotypes; Step 14 significant pathway terms and memberships
BioGRID experimental evidence	Not queried in the reported run

two additional Open Targets-only proteins *GBA1*/*GBA* and *STH*, for a total of 16 protein nodes. These external additions provide reference context; neither was included among the 24 primary strong genes. In particular, the Step 14 gene-level maximum PP_{H4} displayed for *GBA1* was 0.0381 and does not constitute strong GWAS2Mech molecular-colocalisation evidence.

3.11.3 Direct and expanded protein-association networks

The *primary GWAS2Mech-only physical* network comprised 14 nodes and only two high-confidence STRING physical-association edges, connecting *MAPT–SNCA* and *MAPT–NSF*. Eleven of the 14 mapped primary proteins had no edges at the selected threshold in this particular primary physical graph. In the combined seed graph, the inclusion of Open Targets-only *GBA1* added the *SNCA–GBA1* association, yielding 16 nodes and three edges. This combined direct physical graph had 13 connected components and 12 isolates. The difference between the two- and three-edge results reflects the addition of an external comparator, not a change in GWAS2Mech gene evidence.

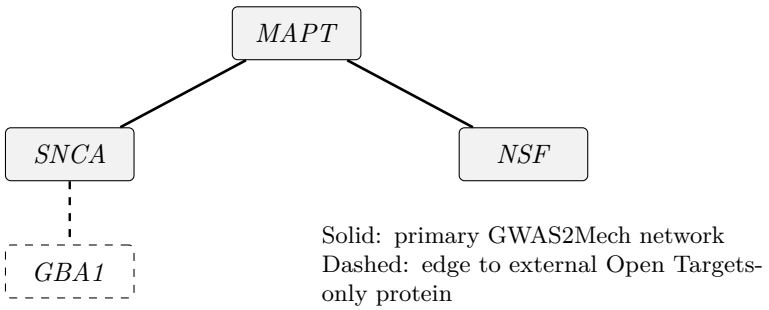
Allowing first-order connector proteins generated an expanded physical-association network with 36 nodes, 26 edges, 12 connected components, 10 isolates and a median STRING edge score of 0.838. The functional association network was denser than its physical counterpart (15 edges among the 16 direct seed proteins; 69 edges among 36 proteins after expansion). These two network types use different STRING definitions and should not be directly interpreted as interchangeable evidence of physical binding. STRING physical edges represent predicted or evidence-integrated physical associations; an edge is not guaranteed to be a direct binary interaction demonstrated experimentally.

3.11.4 Excess protein-association connectivity

Despite the sparseness of the primary physical-only graph, STRING's distinct PPI-enrichment test reported 12 observed protein-association edges, compared with approximately one expected

Table 33: Topology of the four network representations used in workflow step 15. The GWAS2Mech-only primary graph is reported separately because including Open Targets-only seed proteins changes the observed edge count. A dash indicates that the metric was not available in the network summary.

Network	Nodes	Edges	Components	Isolates
Primary GWAS2Mech physical	14	2	12	11
Combined direct physical	16	3	13	12
Expanded physical (+20 connectors)	36	26	12	10
Combined direct functional	16	15	7	6
Expanded functional (+20 connectors)	36	69	–	–



Eleven other primary mapped proteins had no direct high-confidence physical edge and are omitted from this connected-component schematic.

Figure 6: Schematic of the directly connected seed proteins at STRING physical confidence ≥ 0.700. The solid edges (*MAPT–SNCA*, *MAPT–NSF*) are the two primary GWAS2Mech-only associations; the dashed *SNCA–GBA1* association appears only when the external Open Targets seed is included. This simplified diagram is derived from the network summary and does not depict all isolated nodes or the expanded connector network.

edge under its organism-level reference distribution, for the 14 mapped primary proteins ($P = 1.92 \times 10^{-11}$; BH-adjusted $q = 5.76 \times 10^{-11}$). The 19-gene prior-robust set yielded 12 mapped proteins and 11 observed edges, compared with an expected count reported as zero after rounding ($P = 6.26 \times 10^{-12}$; BH-adjusted $q = 3.76 \times 10^{-11}$). The six tests and their corrected results are summarised in Table 34. In these results, the expected edge counts are output-rounded values and should not be used to calculate a precise fold enrichment when zero is printed.

Table 34: STRING PPI-enrichment analysis with BH correction across six tested gene sets. The PPI endpoint measures excess association connectivity in STRING and is distinct from the direct physical graph. Comparison groups retain their upstream Step 14 membership definitions; groups named `CORE_STRONG_OT_*` are not guaranteed to be strict disjoint subsets of the 24-gene `CORE_STRONG` set.

Gene set	Input	Mapped	Edges	P	BH q
`CORE_STRONG`	24	14	12	1.92×10^{-11}	5.76×10^{-11}
`CORE_STRONG_PRIOR_ROBUST`	19	12	11	6.26×10^{-12}	3.76×10^{-11}
`LOCUS_TOP_STRONG`	7	6	0	1.00	1.00
`CORE_STRONG_OT_SUPPORTED`	8	8	3	2.66×10^{-5}	5.32×10^{-5}
`CORE_STRONG_OT_DISCORDANT`	18	8	3	1.20×10^{-3}	1.80×10^{-3}
`OPEN_TARGETS_TOP1`	6	6	2	3.60×10^{-3}	4.32×10^{-3}

Excess STRING connectivity suggests that the subset of mapped candidate proteins is functionally associated within the existing STRING evidence system. It does not validate a disease mechanism independently, because STRING draws on overlapping sources of functional and pathway annotation and because the selected proteins include several genes from the complex chromosome 17 region. Furthermore, the tested genes were prioritised using molecular colocalisation rather than random sampling of the full protein-coding genome; organism-level STRING enrichment is therefore best considered supportive, exploratory network evidence rather than a primary causal statistic.

3.11.5 Network centrality and connector-protein findings

In the *primary* physical graph, *MAPT* had the highest degree (2; connected to *SNCA* and *NSF*); these latter proteins each had degree 1. The other 11 mapped primary proteins were isolated at the chosen physical-association threshold. Thus, the strongest primary result was a small three-protein component rather than a broadly interconnected 14-protein mechanism.

In the expanded *combined-seed* physical graph, *MAPT* became the most connected and highest-betweenness input protein (degree 13; betweenness 0.3613), followed by *SNCA* (degree 8; betweenness 0.2437). External *GBA1* had degree 3. *GAK* and *GCH1*, although isolated in the primary direct physical graph, each gained one predicted physical association after expansion. These changes reflect database-added proteins and should not be interpreted as increased GWAS or colocalisation support for those genes.

Among 20 first-order connector proteins were *DNAJC6*, *SYN1*, *SNCAIP*, *TPPP*, *SNCB*, *GCHFR*, *UGCG*, and *GBA2*. *DNAJC6* and *SYN1* each contacted two seed proteins in the expanded graph; *GCHFR* provided a connection to *GCH1*, while *GAK* was connected through an added protein. Seven connectors were flagged by the script as having a neighbour labelled GWAS2Mech strong and a neighbour labelled Open Targets top-ranked. This count must be

Table 35: Selected nodes in the 36-node expanded physical network. Connector nodes were added algorithmically by STRING and are not part of the primary GWAS2Mech gene set unless separately indicated.

Protein	Node class	Degree	Betweenness	PageRank
MAPT	GWAS2Mech	13	0.36135	0.21261
SNCA	GWAS2Mech + OT	8	0.24370	0.12793
GBA1/GBA	OT-only	3	0.07059	0.04622
DNAJC6	Connector	2	0.03697	0.03868
NSF	GWAS2Mech	2	0.03697	0.03841
SYN1	Connector	2	0	0.03060
GCH1	GWAS2Mech	1	0	0.03636
GAK	GWAS2Mech + OT	1	0	0.02353

interpreted cautiously: a single shared GWAS2Mech/OT seed may satisfy both labels, so the seven flagged connectors do not necessarily bridge two distinct groups of proteins. Connector nodes do not acquire GWAS, fine-mapping or molecular-QTL evidence through their network proximity.

3.11.6 Integration with pathway findings and sensitivity analyses

The Step 15 integration table covered the 90 significant primary pathway terms reported in Step 14, yielding 215 gene–pathway annotation rows. For each pathway, the analysis counted members of the primary gene set and the number of STRING physical edges among those members, and carried forward the previously calculated leave-one-gene-out and leave-one-region-out significance-retention fractions. Representative results are shown in Table 36. The pathway P-values are inherited from Step 14; they are *not* newly generated pathway P-values from the protein network. The Step 14 enrichment used an exploratory annotated-genome background rather than a complete molecular-QTL assay-eligible background, so these terms remain provisional.

Table 36: Selected Step 14 pathway terms contextualised with primary Step 15 physical-network edges. "Physical edges" are the number of edges within the *primary GWAS2Mech* gene members of that pathway. Gene and region retention fractions correspond to separate Step 14 deletion sensitivity tests. No mechanistic direction was inferred.

Pathway/process	Step14 q	Genes	Physical edges	Region LOO	Gene LOO
Secretase involvement in neurodegenerative diseases	0.002152	2	1	4/6	22/24
Synaptic vesicle recycling	0.003226	3	0	4/6	23/24
Synaptic vesicle endocytosis	0.003226	3	0	4/6	23/24
Dopamine biosynthetic process	0.003226	2	0	4/6	22/24
Cellular response to chemical stress	0.003226	4	1	5/6	23/24
Macrophage activation	0.003226	3	1	5/6	23/24
Cellular response to oxidative stress	0.009786	3	1	5/6	23/24
Microglial cell activation	0.009786	2	1	4/6	22/24

The zero primary physical-edge counts for synaptic vesicle recycling, synaptic vesicle endocytosis and dopamine biosynthesis do not invalidate those gene-set annotations: participating proteins may be connected indirectly, have context-specific interactions, or influence a common process without detectable physical binding at this threshold. Conversely, a shared network edge

is not independent experimental validation of a gene-to-pathway causal effect. The gene–pathway export explicitly records pathway direction as `UNKNOWN_NOT_TESTED`.

3.11.7 Interpretation, quality-control caveats, and the next mechanistic stage

The key result of Step 15 was the combination of significant organism-background STRING protein-association enrichment among mapped high-confidence candidates and a relatively sparse primary high-confidence physical-association graph. A core *MAPT–SNCA–NSF* connected component was identifiable directly from primary GWAS2Mech proteins; the *SNCA–GBA1* edge and additional biological connectors emerged in comparator-inclusive or expanded network analyses. The most central nodes of the expanded physical graph, particularly *MAPT* and *SNCA*, are coherent with the neuronal/proteostasis themes of Step 14, but centrality and proximity do not demonstrate causal mediation of Parkinson's disease.

Interpretation requires attention to the following audit findings. First, 10/24 primary candidates were non-protein-coding and could not be assessed as proteins in a standard STRING PPI network. Second, STRING's physical network includes inferred physical associations and was not cross-validated against BioGRID in this run. Third, the output contains a preliminary *LINC02210* symbol-based protein mapping that was subsequently excluded by the biotype audit; protein mapping must always be interpreted jointly with biotype. Fourth, the upstream secondary OT-supported and OT-discordant gene sets are not necessarily disjoint subsets of the 24 primary genes: for example, *TMEM175* was present in an OT-supported secondary set but was not part of the 24 primary genes. Fifth, chromosome 17 contributes many colocalisation candidates and can influence both apparent functional connectivity and pathway enrichment. Finally, neither STRING confidence scores nor the network topology reveal whether a candidate allele increases or decreases a gene's molecular activity or whether perturbing that gene changes a pathway in a particular tissue.

The workflow-step-15 tables therefore provide mechanistic context and an evidence audit rather than establishing causal effector genes. The generated `Step15_Gene_Pathway_Handoff.tsv` preserved gene identifiers, biotypes, pathway memberships, colocalisation evidence, and sensitivity attributes for the subsequent perturbation analysis. Published genetic-perturbation and gene-expression experiments were then used to evaluate gene-to-pathway direction and context where data were available; untested relationships remained unresolved. The network analysis did not support drug or therapeutic-response inference.

3.12 Perturbation analysis of genetically prioritised pathways (workflow step 16)

3.12.1 Rationale and experimental resources

The genome-wide association study (GWAS)-to-mechanism workflow sought to determine whether altering genetically prioritised candidate genes produced measurable downstream transcriptional responses in biological processes implicated by Parkinson's disease (PD) pathway enrichment. The experiment was not a permutation of GWAS variants or randomised disease labels. Rather, it was a secondary analysis of publicly available gene-perturbation single-cell expression datasets,

followed by *gene-set permutation*-based enrichment testing. This distinction matters because an expression-signature enrichment result does not directly estimate the risk conferred by a genetic allele.

The analysis used harmonised public single-cell perturbation resources available through scPerturb [5]. Of 54 inspected `.h5ad` files, 54 were reported as readable, and eight of the 24 primary GWAS2Mech candidate genes had a usable annotated perturbation: *NSF*, *KANSL1*, *GAK*, *MAPT*, *DGKQ*, *IDUA*, *ARL17A*, and *SLC50A1*. These were represented in 22 gene–dataset contrasts, all completing the descriptive analysis. Genes without usable perturbation coverage were not classified as biologically inactive or as negative experimental findings.

3.12.2 Transcriptional contrasts and gene-set permutation analysis

The source analysis was `Step16_Perturbation_Directional_Pathways.py`, version `3.0.0-source-matched-gsea`. For each annotated gene–dataset contrast, cells carrying the target perturbation were compared with the dataset's identified control population. Where relevant metadata were available, control selection accounted for experimental context; nevertheless, equivalence of donor, guide design, perturbation efficiency, and independent biological replication was not established across all datasets. Selected expression matrices were inspected to determine whether their values appeared count-like. Count-like values were transformed as

$$y_{ij} = \log_2 \left(1 + 10^4 \frac{x_{ij}}{\sum_k x_{ik}} \right), \tag{27}$$

where x_{ij} denotes the recorded expression of gene j in cell i. Other nonnegative matrices were analysed on their stored scale. Because expression-scale recognition was heuristic rather than independently checked against every original study's preprocessing, the transformed values should be interpreted cautiously.

For each measured gene j, the descriptive rank was the difference in group means:

$$r_j = \overline{y}_{j,\,\text{perturbed}} - \overline{y}_{j,\,\text{control}}. \tag{28}$$

Genes were ranked by r_j, and the GSEApy implementation of preranked gene-set enrichment analysis was applied to the full available source-specific gene-set libraries [9]. The recorded default configuration used 500 gene-set permutations, random seed 42, and an eligible overlap of 10–500 measured genes per pathway. The actual submitted SLURM options should be retained alongside the publication code release. The test was *not* a biological-replicate-aware test of differential expression: each guide or cell does not automatically represent an independent replicate.

The pathways were identified by their native source identifiers rather than by names alone. The recorded library inventory comprised `GO_Biological_Process_2023` (5,407 library gene sets) and `WikiPathway_2021_Human` (622 library gene sets). Although KEGG support was implemented in the software, a successfully loaded KEGG library was not recorded in this run. Step 14 pathway membership originated from a different enrichment backend and potentially different annotation releases; identity of a GO or WikiPathways term does not guarantee identical gene membership between the two resources.

For every tested pathway, the pipeline recorded a normalised enrichment score (NES), the GSEApy gene-set permutation false-discovery-rate estimate (q_{GSEA}), the leading-edge genes, the source ID, and the experimental cell context. Positive NES denotes enrichment of the pathway's measured genes toward the positive end of the perturbed-minus-control expression ranking; negative NES denotes enrichment toward the negative end. Neither sign establishes increased/decreased biochemical pathway activity or the direction of the PD risk allele. Reported $q_{\mathrm{GSEA}} = 0.000000$ values are software estimates from a finite permutation run, *not* literal zero false-discovery probability. To avoid overstating their precision, these are printed as "0 (reported)" in the tables.

3.12.3 Experimental coverage and source-matched pathway mapping

The 22 completed contrasts yielded 58,627 exploratory pathway-response records. Step 16 reported 1,017 responses overlapping its Step 14 pathway handoff, of which 995 corresponded to GO Biological Process and 22 to WikiPathways; 63 unique Step 14 pathway IDs were experimentally tested, and 69 of the 90 terms provided to Step 16 were exactly mappable to the available library identities. No independent biological-replicate-level FDR was available for these contrasts. The matching results therefore describe observed transcriptional patterns, not replicated causal effects.

Table 37: Recorded Step 16 v3 analysis coverage. A tested pathway or enrichment row is not an independently validated molecular mechanism. The 90 Step 14 terms in the Step 16 handoff must not be confused with the 130 distinct significant pathway terms represented across the broader Step 17 evidence universe.

Measure	Recorded result
Prioritised genes	24
Primary genes with annotated perturbation data	8
Readable local perturbation files	54
Completed gene–dataset contrasts	22
All exploratory pathway-response records	58,627
Step 14 handoff terms assessed for mapping	90
Handoff terms with exact library identity mapping	69
Experimentally tested unique Step 14 IDs (Step 16 scope)	63
Matched pathway-response records in Step 16	1,017
Independent biological-replicate FDR available	No

3.12.4 Exploratory responses passing the gene-set permutation filter

Fourteen pathway–perturbation records satisfied the exploratory software-reported $q_{\mathrm{GSEA}} < 0.05$ filter. These records comprised 12 distinct GO Biological Process terms, two perturbed genes (nine *NSF* responses and five *KANSL1* responses), and six unique gene–dataset contrasts. Critically, the 14 responses did *not* constitute 14 distinct causal mechanisms. Only one response represented the same gene that contributed to the Step 14 enrichment of the corresponding pathway: *NSF* perturbation and Golgi vesicle transport (GO:0048193). The remaining 13 were transcriptional responses of the perturbed gene to a pathway it had not driven in the Step 14 enrichment, and were retained as separate, exploratory cross-layer observations.

Table 38: All 14 reported Step 17 v3 gene-set-permutation-filtered experimental responses ($q_{\text{GSEA}} < 0.05$). *Driver* means that the perturbed gene was also a Step 14 input gene contributing to enrichment of that specific pathway; *response only* means that the gene perturbation was observed to change the pathway-associated transcriptional signature but the perturbed gene was not one of the Step 14 pathway drivers. Step 14 FDR and Step 16 GSEA FDR are different tests. The NSF study/locus shown is an illustrative gene-level genetic association, not a pathway-specific experiment or causal-variant assignment. *KANSL1* study/locus IDs require confirmation from the full Step 12 export.

GO identifier	Biological process	Target	Cell model	NES	GSEA q	Relationship	Step 14 q
GO:0048193	Golgi vesicle transport	*NSF*	RPE1	2.569492	0 (reported)	**Driver**	0.049256
GO:0045333	Cellular respiration	*KANSL1*	Jurkat	-2.294186	0 (reported)	Response only	0.028473
GO:0007005	Mitochondrion organisation	*KANSL1*	K562 essential	-2.044825	0 (reported)	Response only	0.030000
GO:0032271	Regulation of protein polymerisation	*NSF*	RPE1	-1.876480	0.011921	Response only	0.033835
GO:0045333	Cellular respiration	*KANSL1*	HepG2	-1.902643	0.012281	Response only	0.028473
GO:0031110	Regulation of microtubule polymerisation/depolymerisation	*KANSL1*	RPE1	-1.883377	0.016303	Response only	0.021512
GO:0010959	Regulation of metal-ion transport	*NSF*	RPE1	-1.797126	0.021023	Response only	0.028608
GO:1902903	Regulation of supramolecular-fibre organisation	*KANSL1*	RPE1	-1.845246	0.021540	Response only	0.049941
GO:0031110	Regulation of microtubule polymerisation/depolymerisation	*NSF*	RPE1	-1.776954	0.026621	Response only	0.021512
GO:0010038	Response to metal ion	*NSF*	RPE1	-1.770956	0.028407	Response only	0.014243
GO:0031109	Microtubule polymerisation/depolymerisation	*NSF*	RPE1	-1.769217	0.028646	Response only	0.027937
GO:0048545	Response to steroid hormone	*NSF*	RPE1	-1.703010	0.046450	Response only	0.028473
GO:0051402	Neuron apoptotic process	*NSF*	RPE1	-1.693790	0.048481	Response only	0.028280
GO:0019216	Regulation of lipid metabolism	*NSF*	HepG2	1.950397	0.049513	Response only	0.044943

Of the filtered responses, the highest positive NES was observed for *NSF* perturbation in retinal pigment epithelial (RPE1) cells and the Golgi vesicle transport process (NES = +2.569492; GSEA q reported as 0). The same GO term had been enriched among the genetically prioritised genes (Step 14 FDR = 0.049256), with *NSF* and *GAK* as contributing input genes. The reported Step 14 region leave-one-out significance fraction for this pathway was 0.50, underscoring dependence on specific genomic regions. In contrast, *KANSL1* perturbation in Jurkat cells was accompanied by negative enrichment of the cellular-respiration signature (NES = −2.294186; GSEA q reported as 0), and *KANSL1* perturbation in K562 cells was accompanied by negative enrichment of mitochondrion organisation (NES = −2.044825; GSEA q reported as 0). Neither *KANSL1* response is a same-gene Step 14 pathway-driver replication for these terms.

The "neuron apoptotic process" term also passed the exploratory GSEA filter following *NSF* perturbation in RPE1 cells (NES = −1.693790; $q_{\mathrm{GSEA}} = 0.048481$). This result denotes enrichment of expression-ranked genes annotated to neuronal apoptosis; it does *not* establish neuronal apoptosis, neuronal loss, or a protective effect in retinal pigment epithelial cells. Likewise, the "cellular respiration" and "mitochondrion organisation" transcriptional signatures are not direct measurements of oxygen-consumption rate, respiratory-chain activity, or mitochondrial morphology.

The 14 results were selected using FDR values reported within source-specific gene-set analyses for each contrast. They were not subjected to a single additional correction pooling all 58,627 reported pathway records, and overlap among GO terms can induce substantial dependence. Therefore the count of filtered rows must not be interpreted as an experiment-wise family of 14 independently validated discoveries.

3.13 Integration of genetic, pathway, and perturbation evidence (workflow step 17)

3.13.1 Data linkage, mechanism units, and classification

Step 17, version `3.0.0-step16v3-source-id-mechanisms`, integrated molecular association evidence (Step 12), Open Targets comparisons (Step 13), gene-set prioritisation and enrichment (Step 14), protein-association evidence (Step 15), and the experimental pathway responses (Step 16). The underlying analytical unit was *pathway–candidate gene–experimental context*, rather than a claim that each output row represented an independent underlying disease mechanism. Study accession, locus identifier, candidate GWAS and molecular-QTL signal variants, colocalisation posterior, molecular phenotype, pathway driver status, interaction-network evidence, and perturbation response were retained in separate auditable tables and integrated in the mechanism master.

Only pathway records with the same *source and native identifier* were linked; GO Biological Process terms were not equated with Reactome or WikiPathways terms because of similar names. Genetic association, annotated pathway membership, and observed perturbation response were also treated as three distinct propositions. A perturbation of a prioritised gene could yield an expression response for a pathway of which it was *not* a Step 14 contributing gene; such responses were retained, but explicitly labelled `PERTURBATION_RESPONSE_ONLY_NO_STEP14_MEMBERSHIP`.

Conversely, a gene with no matched experimental response remained an unresolved hypothesis rather than a negative result.

The provisional mechanistic-evidence tiers comprised MT1, unresolved gene–pathway relationship; MT2, exploratory experimental pathway response; MT3, replicate-supported experimental pathway response; and MT4, directionally integrated genetic–molecular–experimental evidence satisfying additional consistency requirements. These tiers are rules describing evidence completeness, not statistically calibrated probabilities of gene causality, clinical intervention efficacy, or patient outcome.

For the present dataset, the programme did not establish allele-harmonised GWAS/QTL effect directions, independently validated perturbation target engagement, or replicate-aware experimental FDR. Therefore an MT2 classification does not establish increased or decreased disease risk through the given pathway, and no MT3 or MT4 designations were supported.

3.13.2 Summary of integrated evidence

Step 17 retained 130 unique significant pathway terms across its wider Step 14 enrichment outputs and 395 ranked genes, including 24 primary candidates. Eleven genes had pathway membership in the integrated pathway inventory. The integrated master contained 1,834 candidate gene–pathway–experimental-context records, comprising 1,602 MT2 exploratory records and 232 MT1 unresolved records; no MT3 or MT4 records were assigned. Step 17 reported 95 unique matched pathway identifiers and eight genes represented in experimental records. The difference between the Step 16 handoff count of 63 unique tested Step 14 IDs and the Step 17 total of 95 matched IDs reflects different source/pathway universes and requires a dedicated reconciliation audit rather than being treated as independent additional experiments.

Table 39: Recorded Step 17 v3 evidence inventory. Experimental records are repeated across pathways, genes and cell contexts; none of the row counts is a count of independent validated causal mechanisms.

Measure	Recorded result
Significant unique pathway terms in Step 17	130
Ranked gene identifiers	395
Primary strong prioritised genes	24
Genes with pathway membership in integrated records	11
Candidate pathway–gene–context rows	1,834
MT1 unresolved rows	232
MT2 descriptive experimental rows	1,602
MT3 replicate-supported rows	0
MT4 directionally integrated rows	0
Unique Step 17 matched pathway identifiers	95
Unique experimentally represented genes	8
Filtered $q_{\mathrm{GSEA}} < 0.05$ records (this analysis)	14

3.13.3 Locus-to-gene support for the principal perturbation candidates

The strongest same-gene cross-layer example was *NSF* and Golgi vesicle transport. In the Step 12-derived locus–gene report, *NSF* (ENSG00000073969) had its highest displayed colocalisation

support in GWAS study `GCST90319903`, locus `L006`, where a candidate GWAS signal variant `17:46042921:A:AT` was paired with a molecular-QTL signal variant `17:46255593:C:T` for an *NSF*-related isoform QTL in minor salivary gland ($PP_{H4} = 0.934678$). These two displayed variants are *representatives of GWAS and molecular association signals*, not necessarily the same SNP and not proven to be the shared causal allele. The same study/locus report listed an annotated high-PIP variant `17:46053470:CA:C`, classified as an intronic variant and annotated to *KANSL1*; this locus-level annotation must not be erroneously described as an *NSF* coding mutation or as the variant proven to explain its perturbation response. Additional *NSF* locus-specific molecular associations were available but had lower displayed PP_{H4} (Table 40).

KANSL1 (ENSG00000120071) was independently prioritised among the 24 strong genes. Its workflow-step-14 gene-ranking output recorded a maximum displayed $PP_{H4} = 0.999985$, with an additional molecular-QTL signal labelled as an sQTL in posterior cingulate cortex. Both displayed GWAS and QTL signal identifiers were `17:45969671:T:A`. The available gene-ranking export did not contain the per-study GWAS accession, locus identifier, molecular-effect allele, functional consequence, or direction of the splice change; Table 41 therefore marks these fields as *not verified*. The colocalisation posterior alone cannot establish that *KANSL1* perturbation causes the mitochondrial or respiratory signatures observed in non-neuronal cells.

Table 40: All five available study-specific *NSF* locus–gene records. Variant strings denote candidate GWAS or QTL signal representatives (genome coordinates with alleles), not verified causal variants or harmonised risk alleles. Locus identifiers are study-specific and must be interpreted with their GWAS accession. PP_{H4} quantifies colocalisation support for a shared statistical signal under the adopted model; it is not the posterior probability that *NSF* is the sole causal effector.

GWAS accession	Locus	GWAS signal variant	Molecular-QTL signal variant	PP_{H4}	Modality	Reported molecular context
GCST90319903	L006	17:46042921:A:AT	17:46255593:C:T	0.934678	isoQTL	Minor salivary gland
GCST90476961	L004	17:45854763:C:T	17:45846971:A:C	0.606696	isoQTL	Muscle
GCST90475830	L004	17:45726951:A:G	17:45846971:A:C	0.397978	isoQTL	Muscle
GCST90473307	L008	17:45616172:G:T	17:45606843:C:G	0.203940	eQTL	Testis
GCST90042681	L002	17:46722744:G:A	17:46722744:G:A	0.075780	pQTL	Parahippocampal gyrus

Table 41: Representative gene- and locus-level variant evidence linked to the filtered perturbation results. Values reproduce the supplied Step 12/14 outputs; variant-to-gene assignment, study-specific locus identifier, allelic effect direction, and clinical consequence require independent reconciliation before causal inference. "Region annotation" is not synonymous with gene-specific effect annotation.

Perturbed gene	Genetic context	Candidate variant	Annotation / important caveat
NSF	GCST90319903/L006; isoQTL, minor salivary gland; $PP_{H4} = 0.934678$	GWAS: 17:46042921:A:AT; QTL: 17:46255593:C:T	Signal-level variants only; allelic orientation and disease-risk molecular direction unknown.
NSF	Same GWAS locus; independent regional functional-annotation summary	Top regional PIP: 17:46053470:CA:C	Intronic variant annotated to *KANSL1*, not an established *NSF* causal mutation. Gene-overlapping *NSF* maximum PIP shown as 0 in this particular locus summary.
KANSL1	Strong Step 14 gene ranking; sQTL in posterior cingulate cortex; max $PP_{H4} = 0.999985$	Reported GWAS/QTL representative: 17:45969671:T:A	Gene-level ranking excerpt only; study accession, locus ID, specific splicing event and signed molecular effect require Step 12 verification.

3.13.4 Cross-layer audit of the 14 filtered response records

To provide a direct clinical audit from the pathway-level response to the locus-level association, Table 42 repeats each of the 14 filtered gene–pathway–experiment contexts with the *best available gene-level genetic association*. This does not imply that an experimental gene-set response has been connected to a unique causal variant. A study-specific variant selected in a statistical colocalisation output is not interchangeable with the experimentally edited target, and a gene's strongest reported locus evidence must not be counted repeatedly as independent GWAS replications when several GO terms respond to the same perturbation.

Table 42: Cross-layer audit of all 14 exploratory perturbation records, linking source-native pathway IDs to representative *gene-level* GWAS/QTL evidence. NSF evidence is from the Step 12 study/locus table; KANSL1 is from the Step 14 *gene-level summary only*. The genetic evidence is repeated across multiple pathway-response rows solely to aid traceability. A missing exact study/locus assignment is explicitly marked "not verified". No row demonstrates allele-harmonised causal mediation.

GO ID	Target	Experiment	Genetic accession/locus	Candidate GWAS variant	Candidate QTL variant	Gene best PP_{H4}	Molecular evidence / caveat
GO:0048193	*NSF*	RPE1	GCST90319903/L006	17:46042921:A:AT	17:46255593:C:T	0.934678	isoQTL; minor salivary gland; allelic direction unknown
GO:0045333	*KANSL1*	Jurkat	Not verified	17:45969671:T:A	17:45969671:T:A	0.999985	sQTL; posterior cingulate cortex; per-study locus and sign unresolved
GO:0007005	*KANSL1*	K562	Not verified	17:45969671:T:A	17:45969671:T:A	0.999985	sQTL; posterior cingulate cortex; per-study locus and sign unresolved
GO:0032271	*NSF*	RPE1	GCST90319903/L006	17:46042921:A:AT	17:46255593:C:T	0.934678	isoQTL; minor salivary gland; allelic direction unknown
GO:0045333	*KANSL1*	HepG2	Not verified	17:45969671:T:A	17:45969671:T:A	0.999985	sQTL; posterior cingulate cortex; per-study locus and sign unresolved
GO:0031110	*KANSL1*	RPE1	Not verified	17:45969671:T:A	17:45969671:T:A	0.999985	sQTL; posterior cingulate cortex; per-study locus and sign unresolved
GO:0010959	*NSF*	RPE1	GCST90319903/L006	17:46042921:A:AT	17:46255593:C:T	0.934678	isoQTL; minor salivary gland; allelic direction unknown
GO:1902903	*KANSL1*	RPE1	Not verified	17:45969671:T:A	17:45969671:T:A	0.999985	sQTL; posterior cingulate cortex; per-study locus and sign unresolved
GO:0031110	*NSF*	RPE1	GCST90319903/L006	17:46042921:A:AT	17:46255593:C:T	0.934678	isoQTL; minor salivary gland; allelic direction unknown
GO:0010038	*NSF*	RPE1	GCST90319903/L006	17:46042921:A:AT	17:46255593:C:T	0.934678	isoQTL; minor salivary gland; allelic direction unknown

GO ID	Target	Experiment	Accession/locus	GWAS signal	QTL signal	PP_{H4}	Molecular evidence
GO:0031109	*NSF*	RPE1	GCST90319903/L006	17:46042921:A:AT	17:46255593:C:T	0.934678	isoQTL; minor salivary gland; allelic direction unknown
GO:0048545	*NSF*	RPE1	GCST90319903/L006	17:46042921:A:AT	17:46255593:C:T	0.934678	isoQTL; minor salivary gland; allelic direction unknown
GO:0051402	*NSF*	RPE1	GCST90319903/L006	17:46042921:A:AT	17:46255593:C:T	0.934678	isoQTL; minor salivary gland; allelic direction unknown
GO:0019216	*NSF*	HepG2	GCST90319903/L006	17:46042921:A:AT	17:46255593:C:T	0.934678	isoQTL; minor salivary gland; allelic direction unknown

3.13.5 Criteria for linking genetic and perturbation directions

For each pathway–gene–experiment record, directional integration would require (i) a quality-controlled and allele-harmonised GWAS effect estimate, including phenotype coding and the allele associated with increased disease risk; (ii) a signed molecular-QTL effect on a precisely defined phenotype, such as total expression, isoform usage, or splice-junction abundance; (iii) a verified direction and magnitude of experimental target perturbation; and (iv) a statistically supported downstream response measured in a comparable cell system. The identity of the GWAS and QTL lead variants alone, or a high colocalisation PP_{H4}, does not satisfy these requirements. An isoQTL/sQTL effect cannot be rewritten as an increase or decrease in total gene expression without the appropriate molecular-phenotype definition.

Furthermore, a positive or negative NES is the sign of the *transcriptional ranking* relative to pathway membership. It is not automatically a positive or negative sign for pathway enzymatic activity, clinical disease progression, or drug benefit. Given the absent independent replicate-level differential-expression estimates and unverified molecular/target directions, all filtered responses remained exploratory MT2 evidence. The classification is therefore properly interpreted as *mechanistic hypothesis support* rather than confirmation of a complete causal chain.

3.13.6 Final pathway-to-GWAS and perturbation evidence tables

For clinical interpretability, Tables 43 and 44 provide a two-part audit of the 14 exploratory response records filtered at the software-reported $q_{\mathrm{GSEA}} < 0.05$ threshold. Both tables share the local display identifiers M01–M14; the identifiers denote *gene–pathway–cell-context observations*, not 14 independent causal mechanisms. Table 43 shows the representative genetic association and pathway enrichment for each observation, whereas Table 44 shows its perturbational expression signature and the unresolved links in the proposed causal chain. The Step 12 genetic association is intentionally repeated when one gene has several experimental pathway responses.

Table 43: Final *GWAS-to-pathway evidence table* for all 14 exploratory gene–pathway–experiment observations. Each row shows the Step 14 pathway and the best available *gene-level representative* association. The same NSF study/locus is repeated for distinct experimental responses; these are not independent GWAS discoveries. Blank cells are unavailable or unverified, not negative findings.

ID	Pathway / GO ID	Gene and GWAS study/locus	Candidate signal variants	QTL and annotation	Step14 q
M01	Golgi vesicle transport GO:0048193	*NSF* GCST90319903/L006	GWAS: 17:46042921:A:AT QTL: 17:46255593:C:T	isoQTL, minor salivary gland; $PP_{H4} = 0.934678$. Regional high-PIP annotation: intronic *KANSL1*[a].	0.049256
M02	Cellular respiration GO:0045333	*KANSL1*	GWAS: 17:45969671:T:A[b] QTL: 17:45969671:T:A[b]	sQTL, posterior cingulate cortex; gene-level maximum $PP_{H4} = 0.999985$.	0.028473
M03	Mitochondrion organisation GO:0007005	*KANSL1*	GWAS: 17:45969671:T:A[b] QTL: 17:45969671:T:A[b]	sQTL, posterior cingulate cortex; gene-level maximum $PP_{H4} = 0.999985$.	0.030000
M04	Regulation of protein polymerisation GO:0032271	*NSF* GCST90319903/L006	GWAS: 17:46042921:A:AT QTL: 17:46255593:C:T	isoQTL, minor salivary gland; $PP_{H4} = 0.934678$. Regional high-PIP annotation: intronic *KANSL1*[a].	0.033835
M05	Cellular respiration GO:0045333	*KANSL1*	GWAS: 17:45969671:T:A[b] QTL: 17:45969671:T:A[b]	sQTL, posterior cingulate cortex; gene-level maximum $PP_{H4} = 0.999985$.	0.028473
M06	Regulation of microtubule polymerisation/depolymerisation GO:0031110	*KANSL1*	GWAS: 17:45969671:T:A[b] QTL: 17:45969671:T:A[b]	sQTL, posterior cingulate cortex; gene-level maximum $PP_{H4} = 0.999985$.	0.021512
M07	Regulation of metal-ion transport GO:0010959	*NSF* GCST90319903/L006	GWAS: 17:46042921:A:AT QTL: 17:46255593:C:T	isoQTL, minor salivary gland; $PP_{H4} = 0.934678$. Regional high-PIP annotation: intronic *KANSL1*[a].	0.028608
M08	Regulation of supramolecular-fibre organisation GO:1902903	*KANSL1*	GWAS: 17:45969671:T:A[b] QTL: 17:45969671:T:A[b]	sQTL, posterior cingulate cortex; gene-level maximum $PP_{H4} = 0.999985$.	0.049941
M09	Regulation of microtubule polymerisation/depolymerisation GO:0031110	*NSF* GCST90319903/L006	GWAS: 17:46042921:A:AT QTL: 17:46255593:C:T	isoQTL, minor salivary gland; $PP_{H4} = 0.934678$. Regional high-PIP annotation: intronic *KANSL1*[a].	0.021512
M10	Response to metal ion GO:0010038	*NSF* GCST90319903/L006	GWAS: 17:46042921:A:AT QTL: 17:46255593:C:T	isoQTL, minor salivary gland; $PP_{H4} = 0.934678$. Regional high-PIP annotation: intronic *KANSL1*[a].	0.014243
M11	Microtubule polymerisation/depolymerisation GO:0031109	*NSF* GCST90319903/L006	GWAS: 17:46042921:A:AT QTL: 17:46255593:C:T	isoQTL, minor salivary gland; $PP_{H4} = 0.934678$. Regional high-PIP annotation: intronic *KANSL1*[a].	0.027937
M12	Response to steroid hormone GO:0048545	*NSF* GCST90319903/L006	GWAS: 17:46042921:A:AT QTL: 17:46255593:C:T	isoQTL, minor salivary gland; $PP_{H4} = 0.934678$. Regional high-PIP annotation: intronic *KANSL1*[a].	0.028473

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GWAS–pathway table continued

ID	Pathway / GO ID	Gene; study/locus	GWAS and QTL signals	Molecular QTL / annotation	q_{14}
M13	Neuron apoptotic process GO:0051402	*NSF* GCST90319903/L006	GWAS: 17:46042921:A:AT QTL: 17:46255593:C:T	isoQTL, minor salivary gland; $PP_{H4} = 0.934678$. Regional high-PIP annotation: intronic *KANSL1*[a].	0.028280
M14	Regulation of lipid metabolism GO:0019216	*NSF* GCST90319903/L006	GWAS: 17:46042921:A:AT QTL: 17:46255593:C:T	isoQTL, minor salivary gland; $PP_{H4} = 0.934678$. Regional high-PIP annotation: intronic *KANSL1*[a].	0.044943

[a] The regional high-PIP variant is 17:46053470:CA:C, annotated as intronic to *KANSL1*. It is *not* the same as the displayed NSF GWAS or QTL representative and is not proven to cause the NSF perturbation response. [b] KANSL1 variant identifiers derive from a gene-level signal summary; the associated study accession and locus ID were not independently verified. The reported gene-level maximum PP_{H4} may not be attributable to every displayed molecular association. Step14 q and Step 16 GSEA q are separate statistical measures. PP_{H4} indicates support for a shared association signal under the colocalisation model, not confirmation of a gene-level causal mechanism.

Table 44: Final *mechanism-by-mechanism experimental evidence* for all 14 exploratory Step 16/17 responses. M01–M14 are presentation identifiers, not 14 confirmed molecular mechanisms. "Step14 driver" is Yes only if the experimentally perturbed gene contributed to the genetic enrichment of this exact pathway. All displayed experiments passed the exploratory software-reported GSEA $q < 0.05$ filter, *not* a replicate-aware experimental test. Empty target-effect and risk-direction fields denote unavailable validation evidence.

ID	Pathway / GO term	Gene; model	NES	GSEA q	Step14 driver?	Target effect verified?	Risk direction aligned?	Tier
M01	Golgi vesicle transport GO:0048193	*NSF* RPE1 (RW22)	2.569492	0[c]	Yes			MT2
M02	Cellular respiration GO:0045333	*KANSL1* Jurkat (NO24)	-2.294186	0[c]	No			MT2
M03	Mitochondrion organisation GO:0007005	*KANSL1* K562 essential (RW22)	-2.044825	0[c]	No			MT2
M04	Regulation of protein polymerisation GO:0032271	*NSF* RPE1 (RW22)	-1.876480	0.011921	No			MT2
M05	Cellular respiration GO:0045333	*KANSL1* HepG2 (NO24)	-1.902643	0.012281	No			MT2
M06	Regulation of microtubule polymerisation/depolymerisation GO:0031110	*KANSL1* RPE1 (RW22)	-1.883377	0.016303	No			MT2
M07	Regulation of metal-ion transport GO:0010959	*NSF* RPE1 (RW22)	-1.797126	0.021023	No			MT2
M08	Regulation of supramolecular-fibre organisation GO:1902903	*KANSL1* RPE1 (RW22)	-1.845246	0.021540	No			MT2
M09	Regulation of microtubule polymerisation/depolymerisation GO:0031110	*NSF* RPE1 (RW22)	-1.776954	0.026621	No			MT2
M10	Response to metal ion GO:0010038	*NSF* RPE1 (RW22)	-1.770956	0.028407	No			MT2
M11	Microtubule polymerisation/depolymerisation GO:0031109	*NSF* RPE1 (RW22)	-1.769217	0.028646	No			MT2

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Experimental evidence table continued

ID	Pathway / GO term	Gene; model	NES	GSEA q	Driver?	Target verified?	Risk aligned?	Tier
M12	Response to steroid hormone GO:0048545	*NSF* RPE1 (RW22)	-1.703010	0.046450	No			MT2
M13	Neuron apoptotic process GO:0051402	*NSF* RPE1 (RW22)	-1.693790	0.048481	No			MT2
M14	Regulation of lipid metabolism GO:0019216	*NSF* HepG2 (NO24)	1.950397	0.049513	No			MT2

RW22: Replogle–Weissman 2022 screens; NO24: Nadig–O'Connor 2024 screens. Exact source filenames and target/control cell counts are retained in the Step 16 run audit. All rows had an exploratory response under the software-reported $q < 0.05$ criterion (Yes); independent experimental significance is unknown. For the three entries where software printed $q = 0.000000$, the superscript c denotes a finite-permutation reported zero, not a zero false-discovery probability. Empty cells reflect unavailable verified target engagement and allele-harmonised direction, not negative experimental results. Every row is currently MT2.

Clinical interpretation. The only observation in this filtered set with a *same-gene* connection between Step 14 pathway enrichment and Step 16 perturbation was M01 (*NSF*–Golgi vesicle transport, RPE1). This result is best considered a tractable experimental hypothesis, not evidence that a specific Parkinson's disease risk allele alters Golgi transport in patients. M02–M14 are measurable perturbation-related transcriptional responses associated with genetically enriched pathway *terms*, but the perturbed gene was not among the Step 14 enrichment drivers for that exact term. Independent biological replication and allele-resolved molecular effect direction remain necessary before promotion to a validated mechanistic interpretation.

4 Discussion

4.1 Principal finding and evidential scope

Source-specific pathway-library matching recovered transcriptional perturbation evidence linked to the native identifiers of genetically implicated pathways. Reactome-only matching did not adequately cover the Gene Ontology-dominated pathway inventory from workflow step 14. With source-specific matching, 1,017 pathway-response records overlapped the workflow-step-14 handoff, and the final integration retained 1,602 exploratory gene–pathway–context records. These counts describe evidence integration rather than independent disease mechanisms. Of the 14 filtered responses in Table 38, only one involved a perturbed gene that also drove enrichment of the corresponding pathway, making it the most direct observed bridge between genetic prioritisation and perturbation response.

The study therefore supports three levels of interpretation: first, independently collected genetic/molecular association evidence prioritises candidate genes; second, curated gene-set over-representation associates subsets of those genes with biological processes; and third, perturbation datasets demonstrate whether experimentally changing selected genes is accompanied by a downstream expression signature for a particular process in a particular cell system. The outputs presently do not establish which variant modifies the gene in patients, whether the relevant molecular effect has the same direction as the experimental intervention, or whether the observed pathway signature mediates Parkinsonian neurodegeneration.

4.2 Vesicular transport as the clearest same-gene candidate link

The clearest integrated candidate signal involved *NSF* and Golgi vesicle transport (GO:0048193). *NSF* contributed to the Step 14 gene-set enrichment alongside *GAK* (FDR = 0.049256), and independent perturbation of *NSF* was associated with a strong positive-ranked Golgi vesicle transport signature in RPE1 cells (NES = +2.569492; permutation FDR reported as 0). This observation is consistent with the broader biological relevance of vesicular trafficking to neuronal membrane and protein handling. Its evidential strength lies in the concordance of *gene identity and pathway identity* across genetic annotation and functional perturbation; it does not imply that the GWAS association experimentally reproduces the perturbation or that Golgi trafficking is activated in patients.

Importantly, the Step 12 gene association supporting *NSF* was principally an isoQTL colocalisation signal in minor salivary gland rather than a demonstrated disease-relevant dopaminergic

neuron context. The representative GWAS signal and QTL signal variants differed, and the top regionally annotated high-PIP variant was intronic to *KANSL1*. The broader chromosome 17q21.31 region includes complex haplotypic architecture and long-range linkage disequilibrium; fine-mapped markers and multiple neighbouring gene associations require particularly careful interpretation [19]. Consequently, the RPE1 perturbation cannot be directly assigned to a specific PD risk allele at this locus. The next experimental test should link phased locus-level alleles to verified *NSF* molecular changes and evaluate downstream vesicular-trafficking phenotypes in relevant neural models.

The cross-model comparison further limits broad generalisation. The source-matched Golgi signature for *NSF* was not comparably supported in the reported iPSC-induced neuron experiment (NES $= -0.835779$, $q_{\mathrm{GSEA}} = 0.956533$), and other cell lines displayed heterogeneous signs. This observation does not disprove a vesicle-trafficking mechanism; rather, it demonstrates that experimental model, assay, target engagement and transcriptional context are material to the interpretation. A targeted validation design should consider human dopaminergic neurons and other relevant disease-cell populations, multiple independent perturbation guides, and direct vesicle-trafficking readouts rather than expression-based enrichment alone.

4.3 KANSL1-associated mitochondrial and cytoskeletal responses

A second prominent set of exploratory signatures followed perturbation of *KANSL1*. A strongly negative-ranked cellular-respiration signature was recorded in Jurkat cells (NES $= -2.294186$, GSEA FDR reported as 0) and a separate negative-ranked mitochondrion organisation signature in K562 cells (NES $= -2.044825$, GSEA FDR reported as 0). A cellular-respiration signal was also observed in HepG2 cells (NES $= -1.902643$, $q_{\mathrm{GSEA}} = 0.012281$), providing some cross-model support for a related transcriptional theme. These results suggest that experimentally altering *KANSL1* can affect expression programmes annotated to respiration and mitochondrial biology in the analysed cell lines, but do not directly measure mitochondrial respiration, organelle integrity, mitochondrial membrane potential, or vulnerability to neurodegeneration.

The same-gene genetic pathway link is weaker than for the *NSF*–Golgi example. Although *KANSL1* itself was strongly genetically prioritised ($PP_{H4}^{\max} = 0.999985$), it was *not* among the input genes driving Step 14 enrichment of cellular respiration or mitochondrion organisation in the supplied pathway-driver report. Those Step 14 terms instead involved other genes, including *SNCA* and external-comparator *GBA1*. Thus, the *KANSL1* signatures provide functional responses in a candidate GWAS-prioritised gene, but not a complete independent genetic-to-*KANSL1*-to-mitochondria pathway proof. The Step 14 pathway enrichment was also sensitive to genomic-region exclusion: the reported leave-region-out significance fractions were 0 for cellular respiration and approximately 0.17 for mitochondrion organisation. Their potential relevance should therefore be tested with independent biological replicates and direct functional assays in patient-relevant cells.

Additional negative-ranked *KANSL1* responses involved microtubule dynamics and supramolecular-fibre organisation in RPE1 cells. Together with the mitochondrial terms, these observations motivate experiments probing cytoskeletal organisation and organelle transport. They should not be construed as direct changes to microtubule assembly kinetics or

axonal transport until protein- or imaging-based measurements corroborate the transcriptional observations.

4.4 Other NSF responses: cytoskeleton, ion homeostasis and cell death

The remaining *NSF* responses below the exploratory GSEA threshold were mainly negative-ranked signatures concerning regulation of protein polymerisation, microtubule dynamics, response to metal ions, steroid-hormone response and the neuron-apoptotic-process ontology term in RPE1 cells, together with a positive-ranked regulation-of-lipid-metabolism signature in HepG2 cells. These observations are biologically heterogeneous and often describe overlapping GO annotations. Their coexistence is not equivalent to independent validation of multiple causal pathways, particularly as several nominally different terms can share genes from the same expression-ranked signature.

For example, the negative neuron-apoptotic-process signature (NES $= -1.693790$, $q_{\mathrm{GSEA}} = 0.048481$) is a pathway *expression-enrichment result* measured in retinal pigment epithelial cells; it does not demonstrate an anti-apoptotic effect, neuronal survival benefit, or treatment opportunity in patients. Similarly, terms involving ion transport do not report measured calcium or metal-ion flux, and steroid-hormone-response enrichment does not demonstrate endocrine receptor activation. These are candidates for mechanism-specific phenotyping only after replication and assay validation.

4.5 Pathways without experimental closure

Several biological processes relevant to Parkinson's disease were enriched among the genetically prioritised genes but lacked qualifying same-gene experimental responses in the filtered subset. These included dopamine biosynthesis (represented by *GCH1* and *SNCA* in the workflow-step-14 enrichment inventory), synaptic vesicle recycling (*DGKQ*, *SNCA*, and *GAK*), and glial or microglial activation (including *SNCA* and *MAPT*). The analysed perturbation inventory contained no usable *SNCA* or *GCH1* contrasts. The absence of a functional response therefore reflects experimental coverage and pathway matching rather than evidence against a role in disease.

A clear reporting scheme should distinguish *not assayed*, *assayed but pathway not mapped*, *assayed without exploratory significance*, *exploratory response*, and *replicate-validated response*. None of these categories alone demonstrates directional causal mediation. Data from non-neuronal or transformed cell lines should also remain distinct from neuron-specific Parkinson's disease evidence, even when a queried Gene Ontology term includes a neuron-related label.

4.6 Statistical and genomic limitations

There are five key limitations. First, Step 14 pathway over-representation was performed on a selected and partly related set of GWAS-prioritised genes against an annotated-gene background, rather than an independently established full molecular-QTL-assay-eligible background. The sensitivity of the resulting pathway FDRs to that background should be established before publication-final pathway claims. Second, the 17q21.31 region can contribute correlated ev-

idence for several nearby gene candidates because of structural haplotypes, complex linkage disequilibrium and overlapping molecular QTLs; neither multiple genes at one locus nor multiple related GWAS studies are equivalent to independent mechanistic replications. Third, Step 16's gene-set-permutation FDR does not reflect a full replicate-aware statistical test of the targeted perturbation in biological donors. False discoveries can arise from pseudoreplication in single-cell analyses if biological-level variation is ignored [6]. Fourth, gene-set versions from Step 14 and Step 16 were not guaranteed to have identical membership, even when the native pathway identifier was matched. Fifth, Step 17 lacks verified risk-associated molecular direction and experimentally verified target-effect direction, precluding inference of therapeutic benefit or causal mediation from NES signs alone.

A further reporting nuance is that Step 16 and Step 17 use different collections of upstream pathway terms: 90 handoff terms were available to the Step 16 run, whereas 130 unique significant pathway terms were represented in Step 17. The Step 16 scope yielded 63 tested Step 14 pathway identifiers from its 90-term handoff, while Step 17 matched 95 identifiers within a broader 130-term pathway universe. These figures have different denominators and therefore must not be subtracted or interpreted as independent biological discoveries. Whether every extra Step 17 identifier is traceable to a valid Step 16 experiment remains subject to a source-ID, gene-set-version and run-release audit before submission. This reconciliation is a provenance and integration-quality check, not evidence of either more or fewer independent mechanisms.

4.7 Implications for experimental validation and clinical translation

The principal translational value of these results is the creation of an auditable shortlist for experiments addressing explicitly identified evidence gaps. An appropriate validation programme would first confirm signed GWAS and molecular-QTL effects on harmonised alleles, including the precise molecular phenotype (total expression, isoform usage, splice junction or protein abundance), then test the corresponding increase or decrease in target function rather than assuming that all CRISPRi experiments mimic the risk allele. Cell-type-specific responses should be evaluated in appropriately differentiated human iPSC-derived dopaminergic neurons and, for immune-related mechanisms, relevant microglial or myeloid models. Independent biological donors or experimental replicates, multiple on-target guides, checks for target engagement and rescue experiments should be included where feasible. Replicate-aware expression analysis should precede confirmatory pathway testing; direct vesicle-trafficking, mitochondrial-function, cytoskeletal and neuronal-survival readouts would distinguish pathway-gene-expression signals from physiological mechanism.

The evidence tiers should therefore be described as provisional and contextual. MT1 records identify unresolved pathway–gene hypotheses; MT2 records document descriptive transcriptional responses; MT3 should require appropriate replicate-aware evidence; and MT4 should require independently supported, biologically interpretable concordance among the genetic allele, molecular effect, experimentally changed target and downstream pathway response. None of these categories by itself substitutes for human efficacy or safety evidence.

5 Conclusions

The source-identified integration of workflow steps 16 and 17 recovered exploratory transcriptional responses connected to genetically enriched pathways. The most direct same-gene candidate connection involved *NSF* and Golgi vesicle transport. *KANSL1* perturbations produced respiration-, mitochondrion-, and cytoskeleton-related transcriptional signatures, but *KANSL1* was not a genetic enrichment driver for those specific pathway terms.

These findings narrow a broad GWAS-to-pathway inventory to concrete experimental hypotheses and show why cell context, source identity, and evidence provenance must remain explicit. The present data did not establish risk-allele-to-pathway causal direction, a disease-modifying intervention, or a clinically validated mechanism. GWAS2Mech should therefore be interpreted as a framework for genetically informed mechanistic hypothesis prioritisation with exploratory perturbation evidence.

Data and code availability

The analyses used public summary-level genetic, molecular-QTL, annotation, network, and perturbation resources identified in the resource tables and corresponding citations. The GWAS2Mech source code is available at https://github.com/MuhammadMuneeb007/gwas2mech. A versioned release, immutable commit identifier, and the derived analysis outputs should accompany the manuscript at submission to support exact reproduction.

References

[1] Gao Wang, Abhishek Sarkar, Peter Carbonetto, and Matthew Stephens. A simple new approach to variable selection in regression, with application to genetic fine mapping. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 82(5):1273–1300, 2020. doi: 10.1111/rssb.12388.

[2] Yuxin Zou, Peter Carbonetto, Gao Wang, and Matthew Stephens. Fine-mapping from summary data with the Sum of Single Effects model. *PLOS Genetics*, 18(7):e1010299, 2022. doi: 10.1371/journal.pgen.1010299.

[3] Edward Mountjoy, Ellen M. Schmidt, Miguel Carmona, et al. An open approach to systematically prioritize causal variants and genes at all published human GWAS trait-associated loci. *Nature Genetics*, 53(11):1527–1533, 2021. doi: 10.1038/s41588-021-00945-5.

[4] Open Targets Platform. Open Targets Platform release 26.09. https://blog.opentargets.org/open-targets-platform-26-09-has-been-released/, sep 2026. Release announcement, 25 September 2026; accessed 19 October 2026.

[5] Stefan Peidli, Tessa D. Green, Ciyue Shen, et al. scperturb: harmonized single-cell perturbation data. *Nature Methods*, 21:531–540, 2024. doi: 10.1038/s41592-023-02144-y.

[6] Jordan W. Squair, Matthieu Gautier, Claudia Kathe, et al. Confronting false discoveries in single-cell differential expression. *Nature Communications*, 12, 2021. doi: 10.1038/s41467-021-25960-2.

[7] Damian Szklarczyk, Katerina Nastou, Mikaela Koutrouli, Rebecca Kirsch, Farrokh Mehryary, Radja Hachilif, Dewei Hu, Matteo E. Peluso, Qingyao Huang, Tao Fang, Nadezhda T. Doncheva, Sampo Pyysalo, Peer Bork, Lars J. Jensen, and Christian von Mering. The string database in 2025: protein networks with directionality of regulation. *Nucleic Acids Research*, 53(D1):D730–D737, 2025. doi: 10.1093/nar/gkae1113.

[8] Rose Oughtred, Jennifer Rust, Christie Chang, et al. The biogrid database: A comprehensive biomedical resource of curated protein, genetic, and chemical interactions. *Protein Science*, 30(1):187–200, 2021. doi: 10.1002/pro.3978.

[9] Aravind Subramanian, Pablo Tamayo, Vamsi K. Mootha, et al. Gene set enrichment analysis: a knowledge-based approach for interpreting genome-wide expression profiles. *Proceedings of the National Academy of Sciences*, 102(43):15545–15550, 2005. doi: 10.1073/pnas.0506580102.

[10] Isabel Alfradique-Dunham, Rami Al-Ouran, Rainer von Coelln, et al. Genome-wide association study meta-analysis for parkinson disease motor subtypes. *Neurology Genetics*, 7(2):e557, 2021. doi: 10.1212/NXG.0000000000000557.

[11] Maria Cerezo, Elliot Sollis, Yue Ji, et al. The NHGRI-EBI GWAS Catalog: standards for reusability, sustainability and diversity. *Nucleic Acids Research*, 53(D1):D998–D1005, 2025. doi: 10.1093/nar/gkae1070.

[12] The UK Biobank Whole-Genome Sequencing Consortium. Whole-genome sequencing of 490,640 UK Biobank participants. *Nature*, 645:692–701, 2025. doi: 10.1038/s41586-025-09272-9.

[13] Anurag Verma, Jennifer E. Huffman, Alex Rodriguez, et al. Diversity and scale: Genetic architecture of 2068 traits in the VA Million Veteran Program. *Science*, 385(6706):eadj1182, 2024. doi: 10.1126/science.adj1182.

[14] William McLaren, Laurent Gil, Sarah E. Hunt, Harpreet Singh Riat, Graham R. S. Ritchie, Anja Thormann, Paul Flicek, and Fiona Cunningham. The ensembl variant effect predictor. *Genome Biology*, 17:122, 2016. doi: 10.1186/s13059-016-0974-4.

[15] Philipp Rentzsch, Daniela Witten, Gregory M. Cooper, Jay Shendure, and Martin Kircher. CADD: predicting the deleteriousness of variants throughout the human genome. *Nucleic Acids Research*, 47(D1):D886–D894, 2019. doi: 10.1093/nar/gky1016.

[16] Jun Cheng et al. Accurate proteome-wide missense variant effect prediction with AlphaMissense. *Science*, 381(6664):eadg7492, 2023. doi: 10.1126/science.adg7492.

[17] Melissa J. Landrum et al. ClinVar: improving access to variant interpretations and supporting evidence. *Nucleic Acids Research*, 46(D1):D1062–D1067, 2018. doi: 10.1093/nar/gkx1153.

[18] The ENCODE Project Consortium, Jill E. Moore, et al. Expanded encyclopaedias of DNA elements in the human and mouse genomes. *Nature*, 583:699–710, 2020. doi: 10.1038/s41586-020-2493-4.

[19] Kathryn R. Bowles, Derian A. Pugh, Yiyuan Liu, et al. 17q21.31 sub-haplotypes underlying h1-associated risk for parkinson's disease are associated with lrrc37a/2 expression in astrocytes. *Molecular Neurodegeneration*, 2022. doi: 10.1186/s13024-022-00551-x.